The Dynamics of Marine Ice Sheets and Ice Streams

by

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Abstract

Marine ice sheets are large ice masses that rest on bedrock below sea level and connect to a freely floating ice shelf at the so-called grounding line. We derive, from first principles, two mathematical models for the grounded ice sheet: a weakly lubricated model in which vertical shear stresses provide the dominant resistance to the flow, referred to as a shear-dominated ice sheet; and a well-lubricated model for which rapid sliding overcomes vertical shear, so that extensional stresses dominate, referred to as an extensional ice sheet. We couple both models to an extensional model of the ice shelf.

For the shear-dominated ice sheet, we extend the work of Robison et al. (2010) by incorporating non-Newtonian rheology and the additional geophysical effects of ice accumulation and partial basal slip. The main novelty of this extension is in the force balance at the grounding line, which we derive from first principles and which reduces to Robison et al. (2010) in a special case. In particular, the treatment of the depth-integrated longitudinal stress at the grounding line for the non-Newtonian shear-dominated model requires a new approach that, to the best of our knowledge, has not previously been presented in the published literature. The extensional ice-sheet model follows the framework outlined by Schoof (2007).

We non-dimensionalise the governing equations and solve them numerically in steady state. Exact solutions exist and are found only in limited regions of parameter space, while numerical solutions are used to explore ice-sheet and ice-shelf profiles across a range of basal slip lengths, bed slopes and accumulation rates. The grounding-line position, as calculated by each model, is compared, and power laws are determined in various asymptotic limits. These scalings reveal differences between the two models whilst also revealing intermediate regions of parameter space in which they agree. These results are interpreted geophysically in terms of the physical effects of basal sliding, accumulation and bed slope.

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Chapter 1

Introduction

Ice sheets are vast masses of ice, typically several kilometres deep and hundreds of kilometres long, and are found primarily in Greenland and Antarctica [24]. Typically, they move and deform slowly, progressing roughly one to ten metres per year, as reported by Bingham et al. [4]. Within these sheets are fast-flowing corridors of ice known as ice streams. These move much faster and can progress at up to one thousand metres per year, as reported by Bennett [3]. The variation in velocity is illustrated by the colour-coded ice-velocity map of Antarctica in Figure 1.1, displaying fast-flowing ice streams (purple) among the slowly moving surrounding ice (orange/green).

Despite ice behaving as a solid on small scales, such as the scale of ice cubes in one's freezer, ice is no longer solid on much larger scales, such as that of all of Antarctica. Ice sheets and ice streams experience extremely large, gravity-driven pressure gradients, causing the ice to deform and flow like a viscous fluid. Owing to the relatively small depth of the ice compared with its lateral extent, this motion may be modelled as that of an incompressible thin film of viscous fluid, enabling the use of viscous fluid mechanics to model the flow, as discussed by Greve and Blatter [10]. It is widely acknowledged, for example by Fowler and Larson [7], that ice sheets exhibit flow at extremely low Reynolds numbers of approximately 10^{-12}, and so ice-sheet flow resembles Stokes flow under the force of gravity. The rheology of polycrystalline ice was first explored experimentally by Glen [8], who demonstrated a non-linear power-law relationship between applied stress and strain rate. Specifically, this relationship, now referred to as Glen's flow law, and widely used to model ice sheets is shear thinning with a flow-law exponent of approximately $n = 3$ [8].

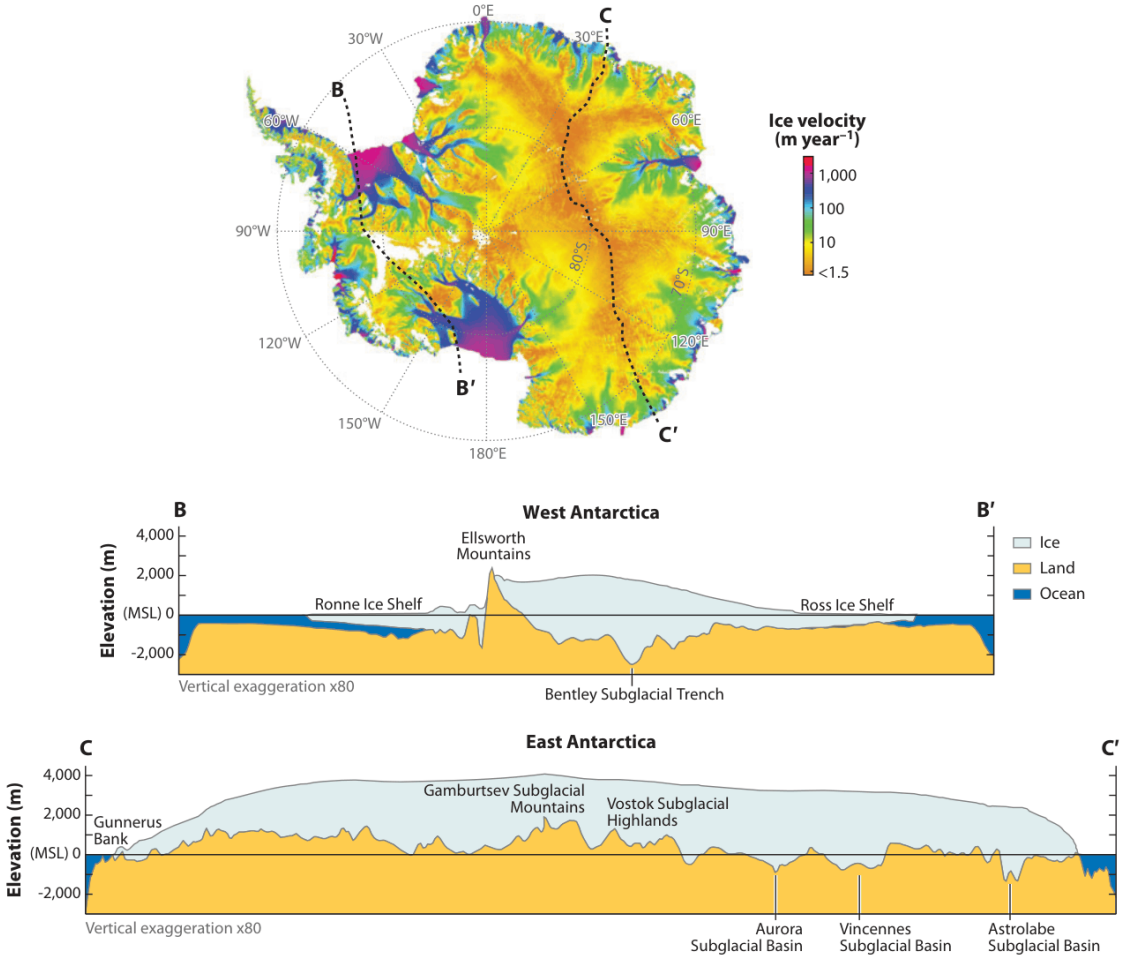


Figure 1.1: Surface velocities across Antarctica (top panel) and cross-sections through West and East Antarctica (middle and bottom panels). West Antarctica is an example of a marine ice sheet, largely grounded below sea level, feeding into the Ronne and Ross ice shelves, while East Antarctica forms a fully continental ice sheet. Reproduced from Schoof and Hewitt [22].

There are two main types of ice sheets: continental and marine ice sheets. Continental ice sheets flow over rigid bedrock for their entire extent [14]. Marine ice sheets, on the other hand, flow across land and extend into the ocean, where they float as freely-floating ice shelves, owing to Archimedes' principle [23]. The point at which the ice sheet lifts off the bedrock, separating the grounded ice sheet from the floating ice shelf, is referred to as the grounding line, or grounding zone [21]. The cross-section of West Antarctica in Figure 1.1 illustrates an example of a marine ice sheet, grounded below sea level for much of its extent before forming the Ross and Ronne ice shelves. East Antarctica, on the other hand, forms a fully continental ice sheet, as shown by the cross-section depicted in the bottom panel of Figure 1.1.

The extent of continental ice sheets is dictated by the accumulation of ice owing to snowfall at high elevations and the loss of ice owing to melting at lower elevations. The mass balance of marine ice sheets additionally depends on the rate of 'calving', or the loss of ice owing to the ice breaking away and forming icebergs at the edge of the ice shelf.

The dominant stress balance within the ice depends on whether it forms part of an ice sheet, an ice shelf or an ice stream. The water beneath an ice shelf and the air above its upper surface exert relatively small tangential stresses on the ice. Consequently, vertical shear stresses are weak, extensional stresses dominate, and ice shelves are typically characterised by plug flow. The resulting equations governing the flow of ice shelves are referred to as extensional-flow equations in the fluid-mechanical literature and as the Shallow Shelf Approximation (SSA) in the glaciological literature [13,22]. The SSA has been successfully used to model both floating ice shelves and well-lubricated ice streams, for which vertical shear is weak and longitudinal stresses are important. MacAyeal [13] developed an early shallow-stream model and applied it to Ice Stream B, while Warburton et al. [25] used the SSA to derive theoretical approximations for the velocity and flux of rapidly sliding ice streams.

Within a continental ice sheet or the grounded portion of a marine ice sheet, the ice rests on solid bedrock and may move over a comparatively thin layer of subglacial till, consisting of an unsorted mixture of clay, silt, sand, and larger rock fragments. This layer can lubricate the base of the ice sheet. The first physical models of this phenomenon were developed by Weertman [26], Nye [15] and Fowler [6], among others, who incorporated the effect of basal sliding into an effective basal boundary condition known as a basal sliding law.

Basal sliding laws relate the basal velocity to the basal shear stress, sometimes involving the effective water pressure at the bed [6, 20]. Difficult access to the underside of ice sheets raised much speculation and debate over the correct form of basal sliding law, resulting in an extensive catalogue of possible sliding laws developed over the past half-century [5]. The simplest of these is the Navier slip law, involving a linear relationship between the basal velocity and basal shear rate [27]. Such a slip law is also relevant to other natural phenomena where slip is common, such as the lotus leaf or man-made superhydrophobic surfaces [18, 27].

Where the basal lubrication is relatively small, the till exerts a significant tangential stress on the base of the ice and, therefore, vertical shear stresses provide the dominant resistance to the flow of the ice sheet. The resulting flow is referred to as a viscous gravity current by the fluid-mechanical community, whereas the geophysical community refers to it as the Shallow Ice Approximation (SIA) [10]. The SIA is commonly used in the geophysical literature, such as in Huybrechts [11] and Greve [9], to model slow-moving grounded ice sheets. In contrast, where basal lubrication is sufficiently large, as in the case of ice streams, basal resistance is reduced and motion due to basal sliding becomes large compared with that due to vertical shearing. In this case, shear stresses become comparatively small, extensional stresses dominate, and the ice approaches plug flow, similar to the ice shelf. As such, the SSA is commonly used to model ice streams as well as ice shelves.

In this thesis, two mathematical models are developed in an idealised two-dimensional geometry to describe the flow of ice sheets under two differing basal sliding conditions: one in which vertical shear stresses are dominant in the ice sheet under partial basal slip, characteristic of moderately lubricated ice sheets, and another in which the sliding of the ice is rapid compared with velocities due to vertical shearing, so that extensional stresses are dominant within the ice sheet, characteristic of well-lubricated ice streams. We will do so by employing the SIA to model the former type of flow, extending Robison et al. [17], and the SSA to model the latter type of flow, following Schoof [21].

The grounding line, separating the grounded ice sheet from the floating ice shelf, involves additional complexity in that multiple viscous stresses are important, thus making a reduced-order model such as the SIA or SSA inapplicable [17, 21]. However, progress has been made by examining the stress balance at the grounding line in an asymptotically small transition region, giving rise to a macroscopic matching condition that can be applied

at the grounding line [12, 17, 21]. Such an approach retains the simplicity of reduced-order models while faithfully representing the grounding-line dynamics, as verified in small-scale laboratory experiments [12, 17]. The effective boundary conditions to be applied at the grounding line depend on the ice-sheet model. For both models, however, the flotation condition, following directly from Archimedes' principle, applies at the grounding line. For the shear-dominated model, we follow Robison et al. [17] by imposing continuity of the depth-integrated longitudinal stress, flux, and ice depth between the ice sheet and the shelf, extending their work to include the additional effects of non-Newtonian rheology, ice accumulation and partial basal slip. For the extensional ice-sheet model, we follow Schoof [21] and impose continuity of the depth-integrated longitudinal stress, flux, and ice depth, while an additional grounding-line flux condition is obtained from the matched-asymptotic analysis of Schoof [21], in which the grounding-line region is treated as a boundary layer.

In the case of a laterally confined ice sheet, experiments by Kowal et al. [12] and Pegler et al. [16] show that the grounding line progresses indefinitely. However, for two-dimensional ice sheets, the grounding line approaches a steady position following an initial advance, as demonstrated experimentally by Robison et al. [17]. Accordingly, steady states are possible in the absence of lateral confinement, and so we restrict our attention to steady two-dimensional configurations throughout this thesis.

The aim of this thesis is to derive and analyse governing equations for two marine ice-sheet models: a shear-dominated model with partial basal slip and an extensional model incorporating rapid basal sliding. The two models are then compared to investigate how the differing basal conditions influence the resulting ice-sheet and ice-shelf profiles and, in particular, the steady grounding-line position. This comparison allows the dependence of the grounding-line position on accumulation, bed slope, basal slip, and other model parameters to be examined.

In Chapter 2, we present a theoretical development of the two models from first principles. In Chapter 3, we non-dimensionalise the governing equations of the two models, compute steady-state solutions numerically, analyse flow profiles, and determine power-law relationships for the grounding-line position in various asymptotic limits. We present our conclusions in Chapter 4.

Chapter 2

Theoretical development

In this chapter, we introduce two models for marine ice sheets and derive their governing equations from first principles.

As mentioned in the introduction, on large scales, ice sheets behave as non-Newtonian fluids in which the dynamic viscosity μ changes with the strain-rate tensor $\mathbf{e} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$ such that $\mu = \mu_0 \left\| \frac{1}{\sqrt{2}}\mathbf{e} \right\|^{\frac{1}{n}-1}$ for some constant $\mu_0 > 0$ and $n > 0$. From geophysical observations, ice is generally modelled with $n = 3$ [8]. Models for the Newtonian case, $n = 1$, and the general non-Newtonian case, $n > 0$, will be presented here. As also mentioned in the introduction, ice sheets exhibit flow at an extremely small Reynolds number $Re = \frac{UL\rho}{\mu} \sim 10^{-12}$, where U and L are typical velocity and length scales, respectively, and ρ is the ice density. Thus, inertial effects can be disregarded, and the dynamics of ice sheets

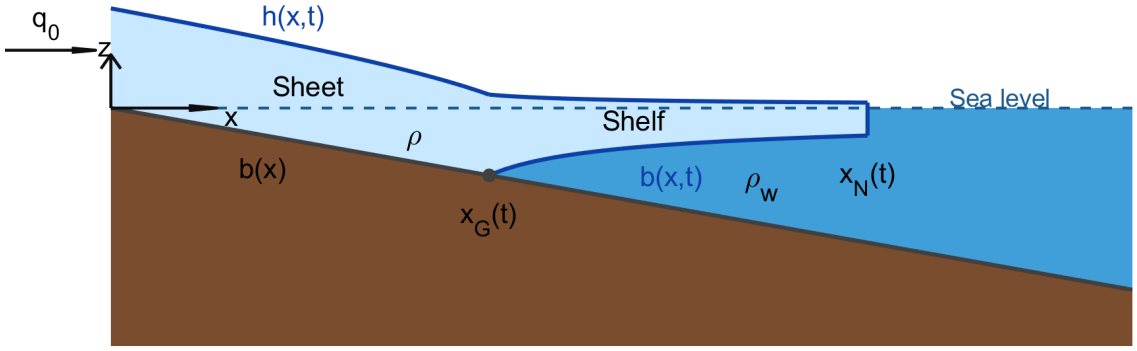


Figure 2.1: Schematic diagram showing an ice sheet with free surface $z = h(x,t)$ flowing down an idealised bed described by $z = b(x) = -\alpha x$ in the region $0 \leq x \leq x_G$, beyond which, in the region $x_G \leq x \leq x_N$, the ice forms a shelf occupying $b(x,t) \leq z \leq h(x,t)$.

and ice streams can be described by the generalised incompressible Stokes equations,

$$\rho\mathbf{g} + \nabla \cdot \boldsymbol{\sigma} = \mathbf{0}, \tag{2.1a}$$

$$\nabla \cdot \mathbf{u} = 0, \tag{2.1b}$$

where $\nabla = \frac{\partial}{\partial x}\mathbf{e}_x + \frac{\partial}{\partial y}\mathbf{e}_y + \frac{\partial}{\partial z}\mathbf{e}_z$, $\mathbf{u}$ describes the ice velocity and ρ the ice density. Here, $\sigma = -pI + 2\mu\mathbf{e}$ is the Cauchy stress tensor, with p the pressure within the ice, and $\mathbf{g} = -g\mathbf{e}_z$, where $g = 9.8\text{ms}^{-2}$, is acceleration due to gravity.

Consider an idealised two-dimensional marine ice sheet shown schematically in Figure 2.1, in which a grounded ice sheet flows down a rigid, steady bed $z = b(x)$, on which a comparatively thin layer of glacial till acts as a lubricant, into a comparatively inviscid ocean of density ρ_w, with $\rho < \rho_w$. The ice forms a free upper surface $z = h(x,t)$ with the much less dense and viscous air above and so the total depth of the ice is given by $H = h - b$. A constant flux, q_0, of ice feeds the ice sheet at the source $x = 0$, where the origin has been chosen such that $b(0)$ corresponds to sea level. At the grounding line $x_G > 0$, the ice lifts off the bedrock, now bounded below by a lower free surface $z = b(x,t)$ between the ice and the ocean, forming a shelf, which floats due to Archimedes' principle and yields the relationship

$$H(x,t) = \frac{\rho_w}{\rho_w - \rho} h(x,t), \tag{2.2}$$

between the upper surface height and ice thickness, as derived in Section 2.2. Ahead of the grounding line, the ice forms a shelf in the region $x_G \leq x \leq x_N$, where x_N is the calving front, or the edge of the shelf.

The depth of the ice is much smaller than its horizontal length, so that $H_0 \ll L_0$, where H_0 and L_0 are typical height and length scales, respectively. Using the scaling laws $\frac{\partial}{\partial x} \sim \frac{1}{L_0}$ and $\frac{\partial}{\partial z} \sim \frac{1}{H_0}$, combined with Equation (2.1b), we find that for a two-dimensional velocity field, $\mathbf{u} = u(x,z,t)\mathbf{e}_x + w(x,z,t)\mathbf{e}_z$, we have $w = O\left(\frac{H_0}{L_0}u\right) \ll u$ and so the flow is approximately horizontal: $\mathbf{u} \approx u\mathbf{e}_x$.

2.1 Mass conservation equation

In this section, the mass conservation equation for the evolution of the ice-stream depth, H, is derived. This is a closure equation, which applies to both types of models of ice sheets considered here.

Writing the continuity equation (2.1b) in components gives

$$\frac{\partial u}{\partial x} + \frac{\partial w}{\partial z} = 0, \tag{2.3}$$

from which we deduce that $\frac{\partial u}{\partial x} = O\left(\frac{1}{L_0}\right)$, and $\frac{\partial w}{\partial z} = O\left(\frac{1}{L_0}\right)$ are of similar scales. Integrating the continuity equation with respect to z yields

$$\int_b^h \frac{\partial u}{\partial x} dz + \int_b^h \frac{\partial w}{\partial z} dz = 0. \tag{2.4}$$

Upon applying the Leibniz rule and the fundamental theorem of calculus, Equation (2.4) becomes

$$\frac{\partial}{\partial x}\int_b^h u\, dz - u\Big|_{z=h}\frac{\partial h}{\partial x} + u\Big|_{z=b}\frac{\partial b}{\partial x} + w(h) - w(b) = 0, \tag{2.5}$$

or

$$\frac{\partial q}{\partial x} + \left[w - u\frac{\partial h}{\partial x}\right]_{z=h} - \left[w - u\frac{\partial b}{\partial x}\right]_{z=b} = 0. \tag{2.6}$$

The kinematic boundary condition at the upper, free surface $z = h$ is given by

$$\frac{D}{Dt}(h - z) = \left(\frac{\partial}{\partial t} + \mathbf{u}\cdot\nabla\right)(h - z) = \frac{\partial h}{\partial t} + u\frac{\partial h}{\partial x} - w = a_h, \tag{2.7}$$

where $\frac{D}{Dt}$ is the material derivative, a_h is the net accumulation, or the difference between the accumulation due to snowfall and the ice melt due to heating from the sun. In our model, it is assumed that melting is comparatively small compared to accumulation at the surface.

Applying a similar boundary condition at $z = b$ yields

$$\frac{D}{Dt}(b - z) = \left(\frac{\partial}{\partial t} + \mathbf{u}\cdot\nabla\right)(b - z) = \frac{\partial b}{\partial t} + u\frac{\partial b}{\partial x} - w = a_b. \tag{2.8}$$

From Equations (2.7) and (2.8), Equation (2.6) may be written as

$$\frac{\partial q}{\partial x} + \left(\frac{\partial h}{\partial t} - a_h\right) - \left(\frac{\partial b}{\partial t} - a_b\right) = 0, \tag{2.9}$$

or,

$$\frac{\partial (h - b)}{\partial t} + \frac{\partial q}{\partial x} = \frac{\partial H}{\partial t} + \frac{\partial q}{\partial x} = a. \tag{2.10}$$

Here, $a = a_h - a_b$ is the net ice accumulation rate. In the case of a steady ice sheet, $\frac{\partial q}{\partial x} = a$ and so the flux is given by

$$q = q_0 + ax. \tag{2.11}$$

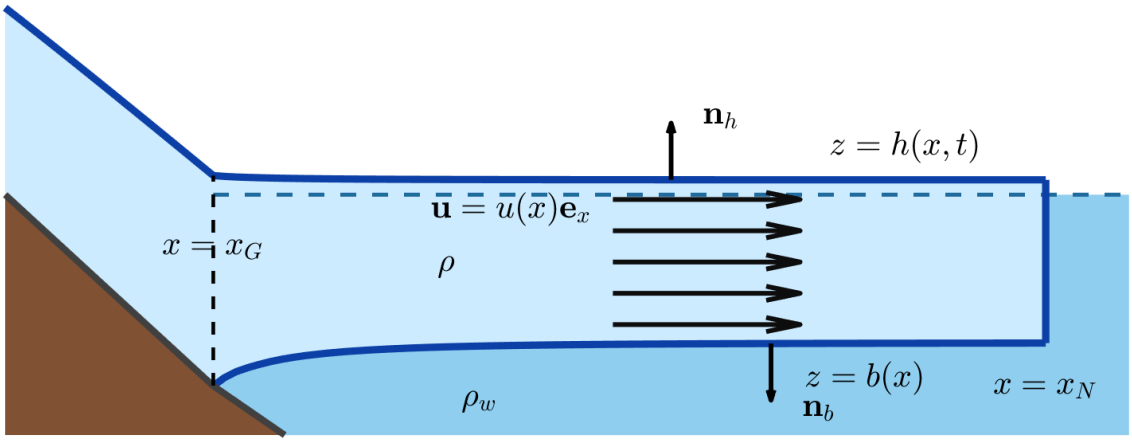


Figure 2.2: Schematic of the floating ice shelf and its connection to the grounded ice sheet at the grounding line $x = x_G$. The upper and lower shelf boundaries are given by $z = h(x,t) = (1 - \rho/\rho_w)\, H$ and $z = b(x) = -(\rho/\rho_w)H$, respectively, as determined by Archimedes' principle. The outward-pointing normals at the upper and lower surfaces are denoted by $\mathbf{n}_h$ and $\mathbf{n}_b$. Horizontal arrows indicate the shelf velocity, $\mathbf{u} = u(x)\mathbf{e}_x$. At the calving front, $x = x_N$, the depth-integrated longitudinal stress is matched to the stress exerted by the ocean, $N = N^{\text{water}}$.

2.2 The ice shelf

In this section, we derive the equations that govern the shape and flow of the ice shelf in the region $x_G \leq x \leq x_N$, illustrated in Figure 2.2. For both models developed for the ice sheet, the ice-shelf equations are identical and follow the same derivation. Our theoretical development closely follows the derivation outlined in Robison et al. [17] for a purely Newtonian shelf and the final result matches that of Schoof [21] for non-Newtonian ice shelves.

Owing to the large difference in the viscosity of the ice compared to the fluids on its upper and lower boundaries, the air and ocean effectively exert no tangential forces on the shelf. Thus, the flow can be assumed to be independent of depth so that $\frac{\partial u}{\partial z} = 0$. From Equation (2.1b), we have $\frac{\partial u}{\partial x} = -\frac{\partial w}{\partial z}$. The strain-rate tensor can thus be rewritten as

$$\mathbf{e} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{1}{2}\left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right) \\ \frac{1}{2}\left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right) & \frac{\partial w}{\partial z} \end{bmatrix} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{1}{2}\left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right) \\ \frac{1}{2}\left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right) & -\frac{\partial u}{\partial x} \end{bmatrix}, \tag{2.12}$$

and so, at leading order,

$$\mathbf{e} = \begin{bmatrix} \frac{\partial u}{\partial x} & 0 \\ 0 & -\frac{\partial u}{\partial x} \end{bmatrix}, \tag{2.13}$$

where $\|\mathbf{e}\| = \sqrt{\sum_{i,j=1}^{2} e_{ij}^2}$. Thus,

$$\|\mathbf{e}\| = \sqrt{2}\left|\frac{\partial u}{\partial x}\right|, \tag{2.14}$$

and so in the non-Newtonian case, the dynamic viscosity is given by $\mu = \mu_0 \left|\frac{\partial u}{\partial x}\right|^{\frac{1}{n}-1}$, which is independent of depth. The stress tensor is thus given by

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{xx} & \sigma_{xz} \\ \sigma_{zx} & \sigma_{zz} \end{bmatrix} = \begin{bmatrix} -p + 2\mu\frac{\partial u}{\partial x} & \mu\left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right) \\ \mu\left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right) & -p - 2\mu\frac{\partial u}{\partial x} \end{bmatrix}, \tag{2.15}$$

the divergence of which at leading order is

$$\nabla \cdot \boldsymbol{\sigma} = \begin{bmatrix} \frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xz}}{\partial z} \\ \frac{\partial \sigma_{zx}}{\partial x} + \frac{\partial \sigma_{zz}}{\partial z} \end{bmatrix} = \begin{bmatrix} \frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xz}}{\partial z} \\ \frac{\partial \sigma_{zz}}{\partial z} \end{bmatrix}. \tag{2.16}$$

Thus, Equation (2.1a) decouples, giving

$$0 = \frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xz}}{\partial z}, \tag{2.17a}$$

$$0 = -\rho g + \frac{\partial \sigma_{zz}}{\partial z}, \tag{2.17b}$$

where Equation (2.17a) and Equation (2.17b) are the x- and z-components, respectively.

Equation (2.17b) can be integrated to obtain

$$\sigma_{zz} = \rho g(z - h), \tag{2.18}$$

where, without loss of generality, atmospheric pressure has been set to zero. From Equation (2.15), $\sigma_{zz} = -p - 2\mu\frac{\partial u}{\partial x}$, which, when combined with Equation (2.18), gives the following expression for pressure

$$p = \rho g(h - z) - 2\mu\frac{\partial u}{\partial x}. \tag{2.19}$$

Applied to Equation (2.15), this yields

$$\sigma_{xx} = -\rho g(h - z) + 4\mu\frac{\partial u}{\partial x}. \tag{2.20}$$

We define the depth-integrated longitudinal stress as

$$N = \int_b^h \sigma_{xx} dz = \int_b^h \left[-\rho g(h - z) + 4\mu\frac{\partial u}{\partial x}\right] dz. \tag{2.21}$$

Upon simplifying,

$$N = -\frac{1}{2}\rho g H^2 + 4\mu H \frac{\partial u}{\partial x}. \tag{2.22}$$

Depth integrating Equation (2.17a) gives rise to

$$\int_b^h \frac{\partial \sigma_{xx}}{\partial x} dz + \int_b^h \frac{\partial \sigma_{xz}}{\partial z} dz = 0, \tag{2.23}$$

and from the Leibniz rule,

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}\int_b^h \sigma_{xx} dz = \int_b^h \frac{\partial \sigma_{xx}}{\partial x} dz + \frac{\partial h}{\partial x}\sigma_{xx}\bigg|_{z=h} - \frac{\partial b}{\partial x}\sigma_{xx}\bigg|_{z=b}. \tag{2.24}$$

Applying the fundamental theorem of calculus, we have

$$\int_b^h \frac{\partial \sigma_{xz}}{\partial z} dz = \sigma_{xz}\bigg|_{z=h} - \sigma_{xz}\bigg|_{z=b}. \tag{2.25}$$

Thus, Equation (2.23) becomes

$$\frac{\partial N}{\partial x} + \left[\sigma_{xz} - \frac{\partial h}{\partial x}\sigma_{xx}\right]_{z=h} - \left[\sigma_{xz} - \frac{\partial b}{\partial x}\sigma_{xx}\right]_{z=b} = 0. \tag{2.26}$$

Consider the stress boundary conditions

$$\boldsymbol{\sigma} \cdot \mathbf{n}_h = \boldsymbol{\sigma}^{\text{air}} \cdot \mathbf{n}_h \quad \text{at} \quad z = h, \tag{2.27a}$$

$$\boldsymbol{\sigma} \cdot \mathbf{n}_b = \boldsymbol{\sigma}^{\text{water}} \cdot \mathbf{n}_b \quad \text{at} \quad z = b, \tag{2.27b}$$

where $\mathbf{n}_h$ and $\mathbf{n}_b$ are the outward unit normals at $z = h$ and $z = b$, respectively. Explicitly,

$$\mathbf{n}_h = \frac{\nabla(z-h)}{|\nabla(z-h)|} = \frac{\left[-\frac{\partial h}{\partial x}, 0, 1\right]^T}{\sqrt{1 + \left(\frac{\partial h}{\partial x}\right)^2}} \quad \text{at} \quad z = h, \tag{2.28a}$$

$$\mathbf{n}_b = \frac{\nabla(b-z)}{|\nabla(b-z)|} = \frac{\left[\frac{\partial b}{\partial x}, 0, -1\right]^T}{\sqrt{1 + \left(\frac{\partial b}{\partial x}\right)^2}} \quad \text{at} \quad z = b. \tag{2.28b}$$

As the viscosity of the air is negligible and atmospheric pressure is set to zero, we have $\boldsymbol{\sigma}^{\text{air}} = 0$, and as the viscosity of the water is negligible, we have $\boldsymbol{\sigma}^{\text{water}} = -p_w \mathbf{I}$, where p_w is the pressure in the ocean just below the shelf. Assuming hydrostatic pressure in the ocean, we have $p_w = -\rho_w g b$ at $z = b$. Therefore, from Equation (2.27a),

$$\begin{bmatrix} \sigma_{xx} & \sigma_{xz} \\ \sigma_{zx} & \sigma_{zz} \end{bmatrix} \begin{bmatrix} -\frac{\partial h}{\partial x} \\ 1 \end{bmatrix} = \begin{bmatrix} \sigma_{xz} - \frac{\partial h}{\partial x}\sigma_{xx} \\ \sigma_{zz} - \frac{\partial h}{\partial x}\sigma_{zx} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \tag{2.29}$$

the x-component of which implies that the boundary term at $z = h$ in Equation (2.26) is equal to zero. From Equation (2.27b),

$$\begin{bmatrix} \sigma_{xx} & \sigma_{xz} \\ \sigma_{zx} & \sigma_{zz} \end{bmatrix} \begin{bmatrix} \frac{\partial b}{\partial x} \\ -1 \end{bmatrix} = \begin{bmatrix} \frac{\partial b}{\partial x}\sigma_{xx} - \sigma_{xz} \\ \frac{\partial b}{\partial x}\sigma_{zx} - \sigma_{zz} \end{bmatrix} = \rho_w g b \begin{bmatrix} \frac{\partial b}{\partial x} \\ -1 \end{bmatrix}, \tag{2.30}$$

the x-component of which is

$$\frac{\partial b}{\partial x}\sigma_{xx} - \sigma_{xz} = \rho_w g b \frac{\partial b}{\partial x}. \tag{2.31}$$

Therefore, Equation (2.26) becomes

$$\frac{\partial N}{\partial x} + \rho_w g b \frac{\partial b}{\partial x} = 0. \tag{2.32}$$

Using Equation (2.22), this can be expressed as

$$\frac{\partial}{\partial x}\left[-\frac{1}{2}\rho g\left(H^2 - \frac{\rho_w}{\rho}b^2\right) + 4\mu H\frac{\partial u}{\partial x}\right] = 0. \tag{2.33}$$

Archimedes' Principle (2.2) states that the weight of the floating object must be equal to the weight of water displaced by the object. Define Δx, Δy as infinitesimally small length components in the x and y directions, respectively. Thus, Archimedes' principle states that for flotation of the ice shelf, $\rho g H \Delta x \Delta y = -\rho_w g b \Delta x \Delta y$, which, upon simplification, yields

$$b = -\frac{\rho}{\rho_w}H, \tag{2.34}$$

or equivalently

$$h = \left(1 - \frac{\rho}{\rho_w}\right)H. \tag{2.35}$$

Applying Archimedes' principle, Equation (2.33) becomes, upon simplifying,

$$\frac{\partial}{\partial x}\left[4\mu H\frac{\partial u}{\partial x} - \frac{1}{2}\rho g' H^2\right] = 0, \tag{2.36}$$

where $g' = \frac{\rho_w - \rho}{\rho_w}g$. Integrating returns

$$4\mu H\frac{\partial u}{\partial x} = \frac{1}{2}\rho g' H^2 + c_1, \tag{2.37}$$

where c_1 is a constant of integration, which is determined by matching the longitudinal stress at the calving front, $x = x_N$, to the water column adjacent to it. Again assuming hydrostatic equilibrium, the pressure in the water is $p_w = -\rho_w g z$, and so $\sigma_{xx}^{\text{water}} = \rho_w g z$. Thus, the depth-integrated longitudinal stress of the water column is

$$N^{\text{water}} = \int_b^0 \sigma_{xx}^{\text{water}} dz = \int_b^0 \rho_w g z dz = -\frac{1}{2}\rho_w g b^2. \tag{2.38}$$

Upon applying Archimedes' principle, this can be expressed as

$$N^{\text{water}} = -\frac{1}{2}\frac{\rho^2}{\rho_w}H^2. \tag{2.39}$$

Upon rewriting and using Equation (2.22),

$$-\frac{1}{2}\rho g H^2 + 4\mu H \frac{\partial u}{\partial x} = -\frac{1}{2}\frac{\rho^2}{\rho_w} g H^2 \quad \text{at} \quad x = x_N. \tag{2.40}$$

Upon simplification and application of Archimedes' principle, this becomes

$$4\mu H \frac{\partial u}{\partial x} = \frac{1}{2}\rho g' H^2 \quad \text{at} \quad x = x_N. \tag{2.41}$$

Hence, $c_1 = 0$ and Equation (2.36) becomes

$$\mu \frac{\partial u}{\partial x} = \frac{\rho g'}{8} H. \tag{2.42}$$

The right-hand side is non-negative, which implies $\frac{\partial u}{\partial x} \geq 0$.

Thus, in the Newtonian case,

$$\frac{\partial u}{\partial x} = \frac{\rho g'}{8\mu} H, \tag{2.43}$$

as in Robison et al. [17]. In the non-Newtonian case,

$$\mu_0 \left| \frac{\partial u}{\partial x} \right|^{\frac{1}{n}-1} \frac{\partial u}{\partial x} = \frac{\rho g'}{8} H. \tag{2.44}$$

Noting that $\frac{\partial u}{\partial x}$ is non-negative, this expression becomes

$$\frac{\partial u}{\partial x} = \left(\frac{\rho g'}{8\mu_0} H \right)^n, \tag{2.45}$$

which agrees with the Newtonian case when $n = 1$, thus generalising the derivation of Robison et al. [17]. Together with the mass conservation equation for constant accumulation, Equation (2.10), the steady-state solution can be found. For this constant-accumulation model, the steady flux equation (2.11) gives rise to a linear flux $q = ax + q_0$, and for plug flow, $q = Hu$. Thus

$$\frac{\partial (Hu)}{\partial x} = \frac{\partial H}{\partial x} u + \frac{\partial u}{\partial x} H = a. \tag{2.46}$$

By expressing the ice velocity in terms of its thickness,

$$u = \frac{q}{H} = \frac{q_0 + ax}{H}, \tag{2.47}$$

we rearrange the extensional flow equation (2.46) and obtain

$$\frac{\partial H}{\partial x} = \frac{aH - H^2 \frac{\partial u}{\partial x}}{q_0 + ax}, \tag{2.48}$$

describing the thickness of the ice shelf in terms of its velocity. Once solved for H, the free surface and base of the ice shelf can be determined by Archimedes' principle

$$h = \left(1 - \frac{\rho}{\rho_w}\right) H, \qquad b = -\frac{\rho}{\rho_w} H. \tag{2.49}$$

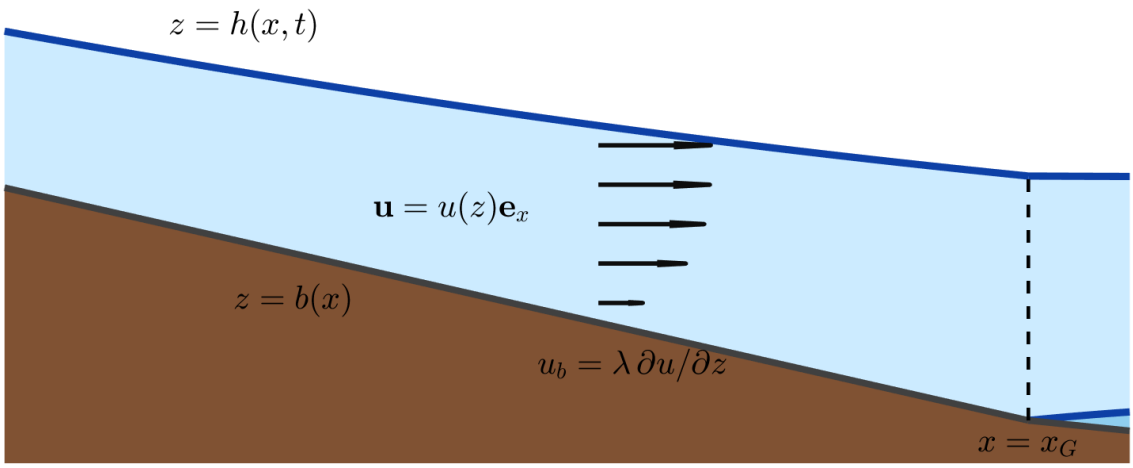


Figure 2.3: Schematic of the grounded ice sheet for the weakly lubricated model, extending slightly beyond the grounding line $x = x_G$. The free surface and bed are given by $z = h(x,t) = H(x,t) - \alpha x$ and $z = b(x) = -\alpha x$, respectively. The horizontal arrows illustrate the depth-dependent velocity profile, $\mathbf{u} = u(z)\mathbf{e}_x$. At the free surface, the tangential stress vanishes, giving $\partial u/\partial z = 0$ at $z = h$, while at the bed, the Navier-slip condition $u = \lambda\, \partial u/\partial z$ is imposed at $z = b$.

2.3 The ice sheet

In this section, we derive the equations describing the shape and flow of the ice sheet in the region $0 \leq x \leq x_G$. Two models will be developed: a model of a moderately lubricated ice sheet, in which the base is partially lubricated and the flow of the ice is resisted predominantly by vertical shear stress, and a model encapsulating rapid basal sliding, in which the flow is resisted predominantly by viscous extensional stress. The former model is an extension of Robison et al. [17] into which we add partial basal slip and accumulation and extend it to non-Newtonian rheology, while the latter model has been explored previously by Schoof [21].

The derivations produce ODEs for the depth, H, while the base and free surface can be obtained from

$$b = -\alpha x, \qquad h = H + b, \tag{2.50}$$

which follow from the model geometry.

2.3.1 Vertical-shear-dominated ice sheets

When the ice sheet is moderately lubricated, the basal slip length is much smaller than the typical ice thickness. As such, vertical shear stress provides the dominant resistance to the flow. We derive governing equations for this class of model below.

For the weakly lubricated model presented here, shown in Figure 2.3, we assume that the basal velocity u_b is directly proportional to the shear rate at the boundary. In the sheet, vertical shear dominates and so, at leading order, the strain-rate tensor is given by

$$\mathbf{e} = \begin{bmatrix} 0 & \frac{1}{2}\frac{\partial u}{\partial z} \\ \frac{1}{2}\frac{\partial u}{\partial z} & 0 \end{bmatrix}. \tag{2.51}$$

We impose the Navier-slip basal boundary condition

$$u_b = \lambda \frac{\partial u}{\partial z}\bigg|_{z=b}, \tag{2.52}$$

where λ is an arbitrary positive constant, known as the slip length. Near $z = b$, the ice velocity increases with distance from the boundary, and so $\frac{\partial u}{\partial z}$ is locally positive.

Under these assumptions, Equation (2.1a) can be written as

$$\mathbf{0} = -\nabla p - \rho g \mathbf{e}_z + \frac{\partial}{\partial z}\left(\mu \frac{\partial u}{\partial z}\right)\mathbf{e}_x, \tag{2.53}$$

the z-component of which is

$$0 = -\frac{\partial p}{\partial z} - \rho g. \tag{2.54}$$

Upon integration we obtain the relationship

$$p = \rho g (h - z). \tag{2.55}$$

The derivation of the governing equations for the sheet for both Newtonian and non-Newtonian cases with partial basal slip will be presented below.

Newtonian flow

Considering the case in which $n = 1$ and so the viscosity of the ice is constant, the x-component of Equation (2.53) is given by

$$0 = -\frac{\partial p}{\partial x} + \mu \frac{\partial^2 u}{\partial z^2} = -\rho g \frac{\partial h}{\partial x} + \mu_0 \frac{\partial^2 u}{\partial z^2}, \tag{2.56}$$

which, upon integration, yields the equations

$$\frac{\partial u}{\partial z} = \left(\frac{\rho g}{\mu}\right)\frac{\partial h}{\partial x}z + A, \tag{2.57}$$

and

$$u = \left(\frac{\rho g}{2\mu}\right)\frac{\partial h}{\partial x}z^2 + Az + B, \tag{2.58}$$

where A and B are arbitrary constants of integration. To determine A, we apply the stress-free condition at $z = h$

$$\frac{\partial u}{\partial z} = 0 \quad \text{at} \quad z = h, \tag{2.59}$$

which yields $A = -\left(\frac{\rho g}{\mu}\right)\frac{\partial h}{\partial x}h$. The constant B can be determined from the basal sliding condition

$$u_b = \lambda\frac{\partial u}{\partial z}\bigg|_{z=b} = -\lambda\left(\frac{\rho g}{\mu}\right)\frac{\partial h}{\partial x}H. \tag{2.60}$$

From Equation (2.58), at $z = b$, we have $u_b = \left(\frac{\rho g}{2\mu}\right)\frac{\partial h}{\partial x}\left(b^2 - 2hb\right) + B$, and so $B = u_b + \left(\frac{\rho g}{2\mu}\right)\frac{\partial h}{\partial x}\left(2hb - b^2\right)$. Thus, the velocity profile is given by

$$u = \left(\frac{\rho g}{2\mu}\right)\frac{\partial h}{\partial x}\left(z^2 - 2hz + 2hb - b^2\right) - \lambda\left[\left(\frac{\rho g}{\mu}\right)\frac{\partial h}{\partial x}H\right], \tag{2.61}$$

or equivalently,

$$u = \left(\frac{\rho g}{2\mu}\right)\frac{\partial h}{\partial x}\left[(h - z)^2 - H^2\right] - \lambda\left[\left(\frac{\rho g}{\mu}\right)\frac{\partial h}{\partial x}H\right]. \tag{2.62}$$

From this expression, the flux, $q = \int_b^h u dz$, through the ice sheet can be determined. Upon integrating and simplifying,

$$q = -\left(\frac{\rho g}{\mu}\right)\frac{\partial h}{\partial x}\left(\frac{1}{3}H^3 + \lambda H^2\right). \tag{2.63}$$

The steady state can be modelled by combining with the steady mass conservation equation with constant accumulation, $\frac{\partial q}{\partial x} = a$, which implies $q = q_0 + ax$. From Equation (2.63),

$$\frac{\partial h}{\partial x} = -\frac{3\mu q}{\rho g\left(H^3 + 3\lambda H^2\right)}. \tag{2.64}$$

Recalling the geometric constraint $H = h - b = h + \alpha x$, we have $\frac{\partial H}{\partial x} = \frac{\partial h}{\partial x} + \alpha$. Thus combining with the expression for q and $\frac{\partial h}{\partial x}$, the depth equation in the ice sheet becomes

$$\frac{\partial H}{\partial x} = \alpha - \frac{3\mu(q_0 + ax)}{\rho g\left(H^3 + 3\lambda H^2\right)}. \tag{2.65}$$

Non-Newtonian flow

The following derivation extends the model presented above to non-Newtonian flow. It follows similar steps to those of Robison et al. [17], but extends their work by incorporating the geophysical effects of ice accumulation and partial basal slip, while also accounting for the additional complexity of non-Newtonian dynamics.

Recall that for non-Newtonian flow, the viscosity is given by $\mu = \mu_0 \left\| \frac{1}{\sqrt{2}}\mathbf{e} \right\|^{\frac{1}{n}-1}$, for some constant μ_0. Where vertical shear stresses dominate, $\mathbf{e}$ is given by (2.51) and so $\mu = \mu_0 \left|\frac{1}{2}\frac{\partial u}{\partial z}\right|^{\frac{1}{n}-1}$. From Equation (2.53), the x-component of the Stokes equations is given by

$$0 = -\frac{\partial p}{\partial x} + \frac{\partial}{\partial z}\left(\mu_0 \left|\frac{1}{2}\frac{\partial u}{\partial z}\right|^{\frac{1}{n}-1} \frac{\partial u}{\partial z}\right). \tag{2.66}$$

Combining with Equation (2.55), we obtain the relation

$$\frac{\partial}{\partial z}\left(\mu_0 \left|\frac{1}{2}\frac{\partial u}{\partial z}\right|^{\frac{1}{n}-1} \frac{\partial u}{\partial z}\right) = \rho g \frac{\partial h}{\partial x}. \tag{2.67}$$

Upon integrating, we find

$$\mu_0 \left|\frac{1}{2}\frac{\partial u}{\partial z}\right|^{\frac{1}{n}-1} \frac{\partial u}{\partial z} = (z + A')\rho g \frac{\partial h}{\partial x}, \tag{2.68}$$

where A' is a constant of integration which is obtained by applying the stress-free boundary condition (2.59), leading to the equation

$$0 = (h + A')\rho g \frac{\partial h}{\partial x}, \tag{2.69}$$

from which we find $A' = -h$. Therefore,

$$\mu_0 \left|\frac{1}{2}\frac{\partial u}{\partial z}\right|^{\frac{1}{n}-1} \frac{\partial u}{\partial z} = (z - h)\rho g \frac{\partial h}{\partial x}. \tag{2.70}$$

Taking the absolute value and rearranging, we find

$$\left|\frac{\partial u}{\partial z}\right| = 2^{1-n}\left|(z-h)\frac{\rho g}{\mu_0}\frac{\partial h}{\partial x}\right|^n. \tag{2.71}$$

Therefore,

$$\frac{\partial u}{\partial z} = 2^{1-n}\gamma\left|(z-h)\frac{\rho g}{\mu_0}\frac{\partial h}{\partial x}\right|^n, \tag{2.72}$$

where

$$\gamma = \begin{cases} 1 & \text{if } \frac{\partial u}{\partial z} > 0 \\ -1 & \text{if } \frac{\partial u}{\partial z} < 0. \end{cases} \tag{2.73}$$

Noting that $z \leq h$, we find

$$\frac{\partial u}{\partial z} = 2^{1-n}\gamma(h-z)^n \left(\frac{\rho g}{\mu_0}\right)^n \left|\frac{\partial h}{\partial x}\right|^n. \tag{2.74}$$

Recalling Equation (2.70), noting that $\mu_0 \left|\frac{1}{2}\frac{\partial u}{\partial z}\right|^{\frac{1}{n}-1} \geq 0$, $z - h \leq 0$ and $\rho g > 0$, it can be determined that $\frac{\partial u}{\partial z}$ and $\frac{\partial h}{\partial x}$ are opposite in sign. Thus $\gamma = -\frac{\partial h}{\partial x}/|\frac{\partial h}{\partial x}|$ and Equation (2.74) can be rewritten as

$$\frac{\partial u}{\partial z} = -2^{1-n}(h-z)^n \left(\frac{\rho g}{\mu_0}\right)^n \frac{\partial h}{\partial x}\left|\frac{\partial h}{\partial x}\right|^{n-1}. \tag{2.75}$$

Integrating again yields

$$u = \frac{2^{1-n}}{n+1}(h-z)^{n+1} \left(\frac{\rho g}{\mu_0}\right)^n \frac{\partial h}{\partial x}\left|\frac{\partial h}{\partial x}\right|^{n-1} + B', \tag{2.76}$$

where $B' = B'(x,t)$ is a constant of integration, which can be determined by applying the basal boundary condition (2.52), leading to the equation

$$u_b = \frac{2^{1-n}}{n+1}(h-b)^{n+1} \left(\frac{\rho g}{\mu_0}\right)^n \frac{\partial h}{\partial x}\left|\frac{\partial h}{\partial x}\right|^{n-1} + B'. \tag{2.77}$$

This implies $B' = u_b - \frac{2^{1-n}}{n+1}(h-b)^{n+1}\left(\frac{\rho g}{\mu_0}\right)^n \frac{\partial h}{\partial x}\left|\frac{\partial h}{\partial x}\right|^{n-1}$, and thus the velocity profile is given by

$$\begin{aligned} u = {} & \frac{2^{1-n}}{n+1}\left[(h-z)^{n+1} - (h-b)^{n+1}\right]\left(\frac{\rho g}{\mu_0}\right)^n \frac{\partial h}{\partial x}\left|\frac{\partial h}{\partial x}\right|^{n-1} \\ & - 2^{1-n}\lambda\left[(h-b)^n \left(\frac{\rho g}{\mu_0}\right)^n \frac{\partial h}{\partial x}\left|\frac{\partial h}{\partial x}\right|^{n-1}\right], \end{aligned} \tag{2.78}$$

which agrees with the Newtonian case when $n = 1$. This profile can be used to determine the flux, which upon integration and simplification is given by

$$q = -\left(\frac{\rho g}{\mu_0}\right)^n \frac{\partial h}{\partial x}\left|\frac{\partial h}{\partial x}\right|^{n-1}\left(\frac{2^{1-n}}{n+2}H^{n+2} + 2^{1-n}\lambda H^{n+1}\right). \tag{2.79}$$

This result again agrees with the Newtonian case when $n = 1$.

We close the problem formulation using the mass conservation equation. Assuming the slope extends downwards, $\frac{\partial h}{\partial x} < 0$, $|\frac{\partial h}{\partial x}| = -\frac{\partial h}{\partial x}$, and so Equation (2.79) can be rearranged to obtain

$$\frac{\partial h}{\partial x} = -\frac{\mu_0}{\rho g}\left(\frac{q}{\frac{2^{1-n}}{n+2}H^{n+2} + 2^{1-n}\lambda H^{n+1}}\right)^{\frac{1}{n}}. \tag{2.80}$$

Recalling $q = q_0 + ax$, and $\frac{\partial H}{\partial x} = \frac{\partial h}{\partial x} + \alpha$, the depth equation for the shear-dominated ice-sheet model with non-Newtonian flow is given by

$$\frac{\partial H}{\partial x} = \alpha - \frac{\mu_0}{\rho g}\left(\frac{q_0 + ax}{\frac{2^{1-n}}{n+2}H^{n+2} + 2^{1-n}\lambda H^{n+1}}\right)^{\frac{1}{n}}. \tag{2.81}$$

This differential equation determines the thickness of the ice sheet and can be integrated numerically, which we will do in a later section.

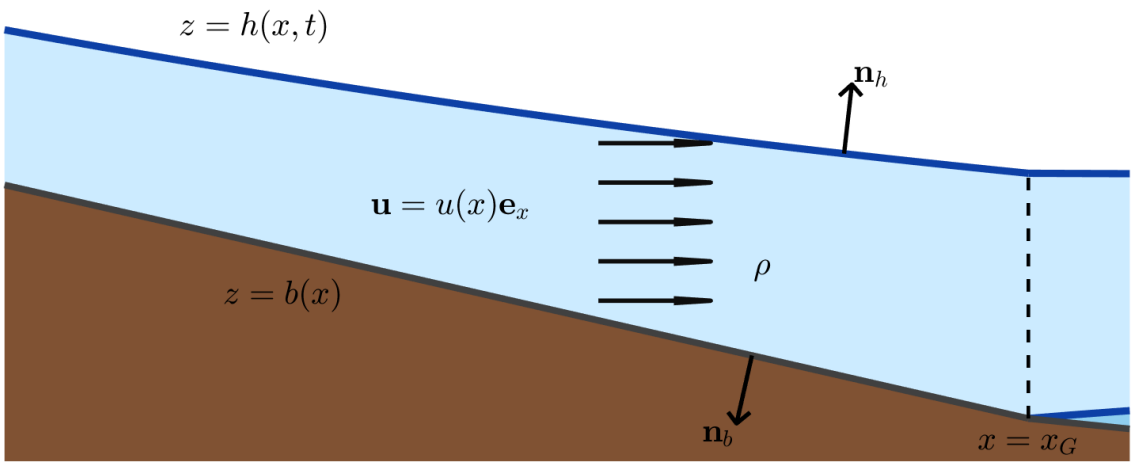


Figure 2.4: Schematic of the grounded ice sheet for the extensional ice-sheet model, extending slightly beyond the grounding line $x = x_G$. The free surface and bed are given by $z = h(x,t) = H(x,t) - \alpha x$ and $z = b(x) = -\alpha x$, respectively. Horizontal arrows indicate the approximately depth-independent velocity, $\mathbf{u} = u(x)\mathbf{e}_x$. The outward-pointing normals to the free surface and bed are denoted by $\mathbf{n}_h$ and $\mathbf{n}_b$. The free surface is stress-free, $\boldsymbol{\sigma}\cdot\mathbf{n}_h = 0$, while the basal traction is prescribed by the sliding law $\boldsymbol{\sigma}\cdot\mathbf{n}_b = -\rho g H\mathbf{n}_b - C|u|^{m-1}u\mathbf{t}$.

2.3.2 Extensional ice sheets undergoing rapid basal sliding

In this section, we outline the derivation of governing equations describing an extensional ice sheet shown schematically in Figure 2.4, as presented by Schoof [21]. Under rapid basal sliding, the bedrock no longer provides sufficient shear stress to deform the ice sheet; the slip length is much greater than the typical ice thickness and so the sheet is instead resisted dominantly by viscous extensional stress. The sheet in this case, therefore, behaves similarly to the shelf, and the derivation closely follows that of the shelf, with the difference being that

$$\boldsymbol{\sigma}\cdot\mathbf{n}_b = \boldsymbol{\sigma}^{\text{till}}\cdot\mathbf{n}_b = -\rho g H\mathbf{n}_b - C|u|^{m-1}u\mathbf{t} \quad \text{at} \quad z = b, \tag{2.82}$$

with C and m positive constants, is the stress at the bed as developed by Schoof [21]. Combined with the outward-pointing normal outlined in Equation (2.28b) and the downstream tangent vector

$$\mathbf{t} = \frac{\begin{bmatrix}1 & \frac{\partial b}{\partial x}\end{bmatrix}^T}{\sqrt{1 + \left(\frac{\partial b}{\partial x}\right)^2}}, \tag{2.83}$$

equation (2.82) becomes

$$\begin{bmatrix} \sigma_{xx} & \sigma_{xz} \\ \sigma_{zx} & \sigma_{zz} \end{bmatrix} \begin{bmatrix} \frac{\partial b}{\partial x} \\ -1 \end{bmatrix} = \begin{bmatrix} \frac{\partial b}{\partial x}\sigma_{xx} - \sigma_{xz} \\ \frac{\partial b}{\partial x}\sigma_{zx} - \sigma_{zz} \end{bmatrix} = -\rho g H \begin{bmatrix} \frac{\partial b}{\partial x} \\ -1 \end{bmatrix} - C|u|^{m-1}u \begin{bmatrix} 1 \\ \frac{\partial b}{\partial x} \end{bmatrix}. \tag{2.84}$$

The x-component of the equation above is $\frac{\partial b}{\partial x}\sigma_{xx} - \sigma_{xz} = -\rho g H \frac{\partial b}{\partial x} - C|u|^{m-1}u$, and so the analogue of Equation (2.32) for the sheet in the case of rapid sliding is

$$\frac{\partial N}{\partial x} = \rho g H \frac{\partial b}{\partial x} + C|u|^{m-1}u. \tag{2.85}$$

From Equation (2.22),

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}\left[4H\mu\frac{\partial u}{\partial x}\right] + \rho g H \frac{\partial b}{\partial x} - \rho g H \frac{\partial h}{\partial x} = \frac{\partial}{\partial x}\left[4H\mu_0 \left|\frac{\partial u}{\partial x}\right|^{\frac{1}{n}-1}\frac{\partial u}{\partial x}\right] + \rho g H \frac{\partial b}{\partial x} - \rho g H \frac{\partial h}{\partial x}. \tag{2.86}$$

Thus Equation (2.85) becomes

$$\frac{\partial}{\partial x}\left[4H\mu_0 \left|\frac{\partial u}{\partial x}\right|^{\frac{1}{n}-1}\frac{\partial u}{\partial x}\right] - C|u|^{m-1}u - \rho g H \frac{\partial h}{\partial x} = 0. \tag{2.87}$$

From Section 3 of Schoof [21], the longitudinal stresses in the sheet are relatively small and so at leading order,

$$C|u|^{m-1}u = -\rho g H \frac{\partial h}{\partial x}. \tag{2.88}$$

Assuming downstream flow, $u > 0$,

$$Cu^m = -\rho g H \frac{\partial h}{\partial x}. \tag{2.89}$$

Upon substitution of the steady flux equation (2.11) and plug flow $u = \frac{q}{H}$, we obtain the relation

$$\frac{\partial h}{\partial x} = -\frac{C}{\rho g H}\left(\frac{q_0 + ax}{H}\right)^m. \tag{2.90}$$

Thus, an explicit equation for the depth of the ice sheet is given by

$$\frac{\partial H}{\partial x} = \alpha - \frac{C}{\rho g H}\left(\frac{q_0 + ax}{H}\right)^m. \tag{2.91}$$

2.3.3 Relationship between the basal sliding parameters

The shear-dominated and extensional models for the grounded ice sheet, although similar in their geometry, differ in their dominant force balance and the basal sliding law. In the shear-dominated model, basal velocity is taken to be linearly proportional to the basal shear rate,

$$u_b = \lambda \left.\frac{\partial u}{\partial z}\right|_{z=b}. \tag{2.92}$$

Here $\lambda > 0$ is the slip length.

For the non-Newtonian case considered here, the basal shear stress is related to the basal shear rate by

$$\tau_b = 2^{1-\frac{1}{n}}\mu_0 \left|\frac{\partial u}{\partial z}\right|^{\frac{1}{n}-1} \frac{\partial u}{\partial z}. \tag{2.93}$$

Thus,

$$\tau_b = 2^{1-\frac{1}{n}}\mu_0 \lambda^{-1/n} |u_b|^{\frac{1}{n}-1} u_b. \tag{2.94}$$

The basal traction law applied in the extensional model for the ice sheet is

$$\tau_b = C|u_b|^{m-1}u_b. \tag{2.95}$$

Comparison of Equations (2.94) and (2.95), therefore, gives

$$m = \frac{1}{n}, \qquad C = 2^{1-\frac{1}{n}}\mu_0\lambda^{-1/n}. \tag{2.96}$$

Thus, the extensional model can be written in terms of the parameters λ and n used in our shear-dominated model.

2.4 The grounding line

In this section, a set of boundary conditions will be developed for both the shear-dominated and extensional models derived in the previous section. These conditions are used to determine the location of the grounding line, x_G, describing the interface between the ice sheet and the ice shelf. To determine these boundary conditions, we assume continuity of depth, flux and depth-integrated longitudinal stress, and a flotation condition derived from Archimedes' principle at the grounding line. The flotation condition applies to both models; it follows from Archimedes' principle discussed in Section 2.2, which, when enforced at the grounding line, returns the condition

$$H_G = \alpha \frac{\rho_w}{\rho} x_G, \tag{2.97}$$

where H_G is the depth of the ice sheet at the grounding line.

Continuity of depth and flux are simply implemented across both models. In fact, flux continuity has already been assumed in the derivation of the shelf and sheet equations. However, there are large differences in the implementation of the continuity of depth-integrated longitudinal stress between the two models. As outlined in previous sections,

the depth-integrated longitudinal stress is given by $N = \int_b^h \sigma_{xx} dz$. Balancing depth-integrated longitudinal stress N between the sheet and the shelf becomes more challenging in the non-Newtonian shear-dominated ice-sheet model due to the non-constant viscosity. Addressing this difficulty requires a new approach which has not previously been presented in the published literature.

As discussed in Robison et al. [17], in geophysical situations, the grounding line is a transition region in which neither the sheet nor shelf equations hold. Across this region, jumps in H, q and N are possible. For our purposes, we can consider the grounding line as a singular surface between the two regions, leading to continuity of these quantities. However, for extensional ice sheets, we consider the flux across this boundary layer as laid out in Schoof [21], where asymptotic matching is applied within this boundary layer to obtain a set of equations to determine the location and depth of the grounding line. We lay this procedure out in what follows.

2.4.1 Shear-dominated ice sheets

Newtonian flow

The following derivation of grounding-line boundary conditions follows the derivation of Robison et al. [17] but with the added geophysical effects of partial basal slip and ice accumulation. The method for matching the depth-integrated longitudinal stress across the grounding line when the sheet is shear-dominated has been developed by Robison et al. [17] for purely Newtonian flows, which we extend to additionally take into account basal sliding and accumulation.

From Equation (2.22), the depth-integrated longitudinal stress in the shelf is given by

$$[N]^{\text{Shelf}} = -\frac{1}{2}\rho g H^2 + 4\mu H \frac{\partial u}{\partial x}. \tag{2.98}$$

In the sheet,

$$[N]^{\text{Sheet}} = -\frac{1}{2}\rho g H^2 + 4\mu \int_b^h \frac{\partial u}{\partial x} dz. \tag{2.99}$$

Applying the Leibniz rule gives rise to

$$[N]^{\text{Sheet}} = -\frac{1}{2}\rho g H_G^2 + 4\mu \left[\frac{\partial q}{\partial x} - u(h)\frac{\partial h}{\partial x} + u(b)\frac{\partial b}{\partial x} \right]. \tag{2.100}$$

The basal boundary condition, given by equation (2.60), provides our sliding velocity at

the base of the sheet. At the surface, the velocity of the ice sheet is

$$u\Big|_{z=h} = -\left(\frac{\rho g}{\mu}\right)\frac{\partial h}{\partial x}\left[\frac{H_G^2}{2} + \lambda H_G\right], \tag{2.101}$$

and so by asymptotic matching, applying continuity of N and H, we obtain

$$\left[\frac{\partial q}{\partial x} + \left(\frac{\rho g}{\mu}\right)\left(\frac{\partial h}{\partial x}\right)^2\left(\frac{H_G^2}{2} + \lambda H_G\right) + \alpha\lambda\left(\frac{\rho g}{\mu}\right)\frac{\partial h}{\partial x}H_G\right]^{\text{Sheet}} = \left[H_G\frac{\partial u}{\partial x}\right]^{\text{Shelf}} \quad \text{at} \quad x = x_G. \tag{2.102}$$

Applying the velocity profile from Equation (2.44), we further simplify to

$$\frac{\partial q}{\partial x} = \frac{\rho g'}{8\mu}H_G^2 - \left(\frac{\rho g}{\mu}\right)\left(\frac{\partial h}{\partial x}\right)^2\left(\frac{H_G^2}{2} + \lambda H_G\right) - \alpha\lambda\left(\frac{\rho g}{\mu}\right)\frac{\partial h}{\partial x}H_G \quad \text{at} \quad x = x_G. \tag{2.103}$$

An equation for the evolution of the grounding-line position can be obtained by considering the differentiated flotation condition,

$$\frac{\partial h}{\partial t} + \frac{\partial h}{\partial x}\frac{\partial x_G}{\partial t} = \left(1 - \frac{\rho}{\rho_w}\right)\left[\frac{\partial H}{\partial t} + \frac{\partial H}{\partial x}\frac{\partial x_G}{\partial t}\right] \quad \text{at} \quad x = x_G. \tag{2.104}$$

Noting that $H = h - b$, we have $\frac{\partial H}{\partial t} = \frac{\partial h}{\partial t}$ and $\frac{\partial H}{\partial x} = \frac{\partial h}{\partial x} - \frac{\partial b}{\partial x}$, which we substitute into the equation above to obtain

$$\frac{\partial h}{\partial t} + \frac{\partial h}{\partial x}\frac{\partial x_G}{\partial t} = -\left(\frac{\rho_w - \rho}{\rho}\right)\frac{\partial b}{\partial x}\frac{\partial x_G}{\partial t} \quad \text{at} \quad x = x_G. \tag{2.105}$$

Applying the mass conservation equation (2.10) and rearranging gives rise to

$$\frac{\partial x_G}{\partial t} = \frac{-a + \frac{\partial q}{\partial x}}{\frac{\partial h}{\partial x} + \left(\frac{\rho_w - \rho}{\rho}\right)\frac{\partial b}{\partial x}} \quad \text{at} \quad x = x_G. \tag{2.106}$$

This expression can be combined with Equation (2.103) to obtain an equation for the grounding-line evolution. In steady state, the thickness H_G at the grounding line is determined by solving

$$\frac{\partial q}{\partial x} - a = \frac{\rho g'}{8\mu}H_G^2 - \left(\frac{\rho g}{\mu}\right)\left(\frac{\partial h}{\partial x}\right)^2\left(\frac{H_G^2}{2} + \lambda H_G\right) - \alpha\lambda\left(\frac{\rho g}{\mu}\right)\frac{\partial h}{\partial x}H_G - a = 0 \quad \text{at} \quad x = x_G. \tag{2.107}$$

Upon applying Equation (2.63), with $q = q_0 + ax$ from the mass conservation equation, we find

$$\frac{\partial h}{\partial x} = -\frac{\mu}{\rho g}\frac{q_0 + ax_G}{\frac{1}{3}H_G^3 + \lambda H_G^2} \quad \text{at} \quad x = x_G. \tag{2.108}$$

Upon applying the mass conservation equation (2.10) to Equation (2.103), we obtain

$$0 = \frac{\rho g'}{8\mu}H_G^2 - \left(\frac{\rho g}{\mu}\right)\left(\frac{\partial h}{\partial x}\right)^2\left(\frac{H_G^2}{2} + \lambda H_G\right) - \alpha\lambda\left(\frac{\rho g}{\mu}\right)\frac{\partial h}{\partial x}H_G - a \quad \text{at} \quad x = x_G. \tag{2.109}$$

This can be solved numerically with Equation (2.108) to obtain H_G. The grounding-line position x_G can thus be found by applying the flotation condition combined with the bed geometry, giving

$$x_G = \frac{\rho}{\rho_w \alpha} H_G. \tag{2.110}$$

Non-Newtonian flow

The following derivation of the grounding-line boundary conditions for non-Newtonian flow applies continuity of the same quantities as Robison et al. [17], adding again the geophysical effects of ice accumulation and partial basal slip. Difficulties arise with the evaluation of depth-integrated longitudinal stress for the non-Newtonian ice sheet due to the non-constant viscosity. To overcome these difficulties, a new approach is developed. Although previous works, such as Sayag and Worster [19] and Greve and Blatter [10], have derived models for shear-thinning gravity currents, to the best of our knowledge no published work has coupled such a model to an ice shelf and derived the corresponding grounding-line equations. In this respect, the derivation presented here is novel.

When the flow is non-Newtonian, the depth-integrated longitudinal stress of the ice sheet is more difficult to evaluate due to the non-constant viscosity. The depth-integrated longitudinal stress in the sheet is given by

$$[N]^{\text{Sheet}} = -\frac{1}{2}\rho g H_G^2 + 4\mu_0 \int_b^h \left|\frac{1}{2}\frac{\partial u}{\partial z}\right|^{\frac{1}{n}-1} \frac{\partial u}{\partial x} dz. \tag{2.111}$$

Upon defining the integral

$$I = \int_b^h \left|\frac{1}{2}\frac{\partial u}{\partial z}\right|^{\frac{1}{n}-1} \frac{\partial u}{\partial x} dz, \tag{2.112}$$

it follows from Equation (2.75) that

$$\left|\frac{1}{2}\frac{\partial u}{\partial z}\right|^{\frac{1}{n}-1} = 2^{n-1}(h-z)^{1-n}\left(\frac{\rho g}{\mu_0}\right)^{1-n}\left|\frac{\partial h}{\partial x}\right|^{1-n}. \tag{2.113}$$

The x-derivative of the velocity is

$$\begin{aligned}\frac{\partial u}{\partial x} = {} & 2^{1-n}\left(\frac{\rho g}{\mu_0}\right)^n \frac{\partial h}{\partial x}\left|\frac{\partial h}{\partial x}\right|^{n-1}\left[(h-z)^n\frac{\partial h}{\partial x} - H^n\frac{\partial H}{\partial x} - n\lambda H^{n-1}\frac{\partial H}{\partial x}\right] \\ & + 2^{1-n} n\left(\frac{\rho g}{\mu_0}\right)^n \left|\frac{\partial h}{\partial x}\right|^{n-1}\frac{\partial^2 h}{\partial x^2}\left[\frac{(h-z)^{n+1} - H^{n+1}}{n+1} - \lambda H^n\right].\end{aligned} \tag{2.114}$$

Substituting these expressions into I and simplifying gives

$$I = \frac{\rho g}{\mu_0}\left[\left(\frac{\partial h}{\partial x}\right)^2 \int_b^h (h-z)\,dz + \frac{n}{n+1}\frac{\partial^2 h}{\partial x^2}\int_b^h (h-z)^2\,dz \right. \\ \left. - \left[\frac{\partial h}{\partial x}\frac{\partial H}{\partial x}\left(H^n + n\lambda H^{n-1}\right) + n\frac{\partial^2 h}{\partial x^2}\left(\frac{H^{n+1}}{n+1} + \lambda H^n\right)\right]\int_b^h (h-z)^{1-n}, dz\right]. \quad (2.115)$$

Issues arise with the term

$$\int_b^h (h-z)^{1-n} dz, \quad (2.116)$$

which diverges for $n \geq 2$. For modelling ice, we are interested in $n = 3$ and so evaluating the depth-integrated longitudinal stress in this way is not useful for the geophysical scenario which we wish to model. Other issues arise if we instead replace the reduced stress tensor by the full stress tensor $\|\mathbf{e}\| = \sqrt{2\left(\frac{\partial u}{\partial x}\right)^2 + \frac{1}{2}\left(\frac{\partial u}{\partial z}\right)^2}$, from which we deduce the viscosity $\mu = \mu_0\left(\left(\frac{\partial u}{\partial x}\right)^2 + \frac{1}{4}\left(\frac{\partial u}{\partial z}\right)^2\right)^{\frac{1-n}{2n}}$. Evaluating the depth-integrated longitudinal stress with this model leads to a non-elementary integral.

Analytical progress can be made, however, when we assume that extensional stress dominates near the grounding line. Such a scenario is geophysically relevant as subglacial till has been found to accumulate in the grounding zone of ice sheets as reported by Alley [1] and Anandakrishnan et al. [2]. In turn, this increases subglacial lubrication significantly in the grounding zone in comparison to that of the bulk of the ice sheet. In this scenario, the ice viscosity simplifies to $\mu = \mu_0 |\frac{\partial u}{\partial x}|^{\frac{1}{n}-1}$, and the depth-integrated longitudinal stress is

$$[N]^{\text{Sheet}} = -\frac{1}{2}\rho g H_G^2 + 4\mu_0 \int_b^h \left|\frac{\partial u}{\partial x}\right|^{\frac{1}{n}-1}\frac{\partial u}{\partial x} dz. \quad (2.117)$$

We now focus on the final term, which we define in terms of the integral

$$J = \int_b^h \left|\frac{\partial u}{\partial x}\right|^{\frac{1}{n}-1}\frac{\partial u}{\partial x} dz. \quad (2.118)$$

Upon substituting the expression for the velocity field into the integral, we find that the result is a non-elementary integral and so cannot be evaluated exactly. To make analytical progress, we define the average velocity gradient

$$\overline{u_x} = \frac{1}{H_G}\int_b^h \frac{\partial u}{\partial x} dz, \quad (2.119)$$

which we evaluate by applying the Leibniz rule, which gives us

$$\overline{u_x} = \frac{1}{H_G}\left[\frac{\partial q}{\partial x} - u(h)\frac{\partial h}{\partial x} + u(b)\frac{\partial b}{\partial x}\right]. \quad (2.120)$$

Making the assumption that $J \approx H_G|\overline{u_x}|^{\frac{1}{n}-1}\overline{u_x}$, continuity of depth-integrated longitudinal stress N implies

$$\frac{\partial q}{\partial x} - u(h)\frac{\partial h}{\partial x} + u(b)\frac{\partial b}{\partial x} = \left(\frac{\rho g'}{8\mu_0}\right)^n H_G^{n+1}. \tag{2.121}$$

Applying the velocity at the boundaries and the steady-state mass conservation equation, we have

$$a + \left(\frac{\rho g}{\mu_0}\right)^n \frac{\partial h}{\partial x}\left|\frac{\partial h}{\partial x}\right|^{n-1}\left[\frac{2^{1-n}H_G^{n+1}}{n+1}\frac{\partial h}{\partial x} + 2^{1-n}\lambda H_G^n\left(\frac{\partial h}{\partial x} + \alpha\right)\right] - \left(\frac{\rho g'}{8\mu_0}\right)^n H_G^{n+1} = 0, \tag{2.122}$$

at $x = x_G$.

From Equation (2.80), at the grounding line,

$$\frac{\partial h}{\partial x} = -\frac{\mu_0}{\rho g}\left[\frac{q_0 + a x_G}{\frac{2^{1-n}}{n+2}H_G^{n+2} + \lambda H_G^{n+1}}\right]^{\frac{1}{n}} \quad \text{at} \quad x = x_G. \tag{2.123}$$

The steady grounding-line height can be determined by substitution of this equation for $\frac{\partial h}{\partial x}$ into Equation (2.122) and numerically solving for H_G. The grounding-line position x_G can be determined by applying the flotation condition (2.110).

2.4.2 Extensional ice sheets

We now fill in the theoretical details behind the development of the matching conditions of Schoof [21] at the grounding line of extensional ice sheets. For extensional ice sheets, the depth-integrated longitudinal stress in the sheet is similar to that of the shelf, giving rise to

$$[N]^{\text{Sheet}} = -\frac{1}{2}\rho g H_G^2 + 4\mu H_G\frac{\partial u}{\partial x}. \tag{2.124}$$

Matching N at the grounding line and applying continuity of thickness H provides the boundary condition

$$\left[\frac{\partial u}{\partial x}\right]^{\text{Sheet}} = \left[\frac{\partial u}{\partial x}\right]^{\text{Shelf}} \quad \text{at} \quad x = x_G. \tag{2.125}$$

As there is plug flow within both the sheet and the shelf, flux continuity,

$$[Hu]^{\text{sheet}} = [Hu]^{\text{shelf}} \quad \text{at} \quad x = x_G, \tag{2.126}$$

combined with depth continuity, implies

$$[u]^{\text{sheet}} = [u]^{\text{shelf}} \quad \text{at} \quad x = x_G. \tag{2.127}$$

Both of these conditions agree with those provided by Schoof [21].

The position of the grounding line can be determined through asymptotic matching as described in Section 3 of Schoof [21]. By matching the inner and outer solutions at $x = x_G$, the relation

$$q_G = K H_G^{\frac{m+n+3}{m+1}}, \tag{2.128}$$

where

$$K = \left[\frac{\rho g}{C} \left(\frac{\rho g'}{8\mu_0} \right)^n \right]^{1/(m+1)}, \tag{2.129}$$

is obtained.

This is combined with the flotation condition and flux equation (2.11) to obtain

$$q_0 + \frac{a\rho}{\rho_w \alpha} H_G = K H_G^{\frac{m+n+3}{m+1}}. \tag{2.130}$$

This can be solved for H_G, from which the grounding-line position can be determined in combination with the flotation condition, which yields

$$x_G = \frac{\rho}{\rho_w \alpha} H_G. \tag{2.131}$$

2.5 Summary of equations

In what follows, we provide a summary of the governing equations and boundary conditions describing the coupled ice sheet and ice shelf for both shear-dominated and extensional ice sheets.

2.5.1 Shear-dominated ice sheets

Newtonian flow

In the ice sheet, for $0 \leq x \leq x_G$,

$$\frac{\partial H}{\partial x} = \alpha - \frac{3\mu(q_0 + ax)}{\rho g \left(H^3 + 3\lambda H^2\right)}, \tag{2.132a}$$

$$b = -\alpha x, \qquad h = H + b, \tag{2.132b}$$

$$u = \frac{\rho g}{2\mu} \left[(h-z)^2 - (h-b)^2 \right] \frac{\partial h}{\partial x} - \lambda \frac{\rho g}{\mu} (h-b) \frac{\partial h}{\partial x}. \tag{2.132c}$$

At the grounding line, $x = x_G$,

$$0 = \frac{\rho g'}{8\mu} H_G^2 - \frac{\rho g}{\mu}\left(\frac{\partial h}{\partial x}\right)^2 \left(\frac{H_G^2}{2} + \lambda H_G\right) - \alpha\lambda \frac{\rho g}{\mu}\frac{\partial h}{\partial x} H_G - a, \tag{2.133a}$$

$$\frac{\partial h}{\partial x} = -\frac{\mu}{\rho g}\frac{q_0 + a x_G}{\frac{1}{3}H_G^3 + \lambda H_G^2}, \tag{2.133b}$$

$$x_G = \frac{\rho}{\rho_w \alpha} H_G. \tag{2.133c}$$

In the ice shelf, for $x_G \leq x \leq x_N$,

$$\frac{\partial H}{\partial x} = \frac{a H - \frac{\rho g'}{8\mu} H^3}{q_0 + a x}, \tag{2.134a}$$

$$b = -\frac{\rho}{\rho_w} H, \qquad h = \left(1 - \frac{\rho}{\rho_w}\right) H, \tag{2.134b}$$

$$u = \frac{q_0 + a x}{H}. \tag{2.134c}$$

This system of governing equations, excluding the additional effects of ice accumulation and basal sliding, was previously derived by Robison et al. [17].

Non-Newtonian flow

In the ice sheet, for $0 \leq x \leq x_G$,

$$\frac{\partial H}{\partial x} = \alpha - \frac{\mu_0}{\rho g}\left[\frac{q_0 + a x}{\frac{2^{1-n}}{n+2} H^{n+2} + 2^{1-n}\lambda H^{n+1}}\right]^{1/n}, \tag{2.135a}$$

$$b = -\alpha x, \qquad h = H + b, \tag{2.135b}$$

$$u = \left(\frac{\rho g}{\mu_0}\right)^n \frac{\partial h}{\partial x}\left|\frac{\partial h}{\partial x}\right|^{n-1}\left[\frac{2^{1-n}}{n+1}\left((h-z)^{n+1} - H^{n+1}\right) - 2^{1-n}\lambda H^n\right]. \tag{2.135c}$$

At the grounding line, $x = x_G$,

$$0 = a + \left(\frac{\rho g}{\mu_0}\right)^n \frac{\partial h}{\partial x}\left|\frac{\partial h}{\partial x}\right|^{n-1}\left[\frac{2^{1-n}}{n+1} H_G^{n+1}\frac{\partial h}{\partial x} + 2^{1-n}\lambda H_G^n\left(\frac{\partial h}{\partial x} + \alpha\right)\right] - \left(\frac{\rho g'}{8\mu_0}\right)^n H_G^{n+1}, \tag{2.136a}$$

$$\frac{\partial h}{\partial x} = -\frac{\mu_0}{\rho g}\left[\frac{q_0 + a x_G}{\frac{2^{1-n}}{n+2} H_G^{n+2} + 2^{1-n}\lambda H_G^{n+1}}\right]^{1/n}, \tag{2.136b}$$

$$x_G = \frac{\rho}{\rho_w \alpha} H_G. \tag{2.136c}$$

In the ice shelf, for $x_G \leq x \leq x_N$,

$$\frac{\partial H}{\partial x} = \frac{aH - H^2\left(\frac{\rho g'}{8\mu_0}H\right)^n}{q_0 + ax}, \tag{2.137a}$$

$$b = -\frac{\rho}{\rho_w}H, \qquad h = \left(1 - \frac{\rho}{\rho_w}\right)H, \tag{2.137b}$$

$$u = \frac{q_0 + ax}{H}. \tag{2.137c}$$

This system of equations is an extension of Robison et al. [17]. Parts of them have been derived previously in other works; Equations (2.135) have been derived by Sayag & Worster [19] in the context of viscous gravity currents of power-law fluids spreading over a rigid substrate, while Equations (2.137) have been used by Schoof [21] to describe the flow of freely floating ice shelves. The grounding-line conditions (2.136), coupling a shear-dominated ice sheet to an extensional ice shelf are new to the best of our knowledge and reduce to those of Robison et al. [17] when the flow is Newtonian ($n = 1$) and when there is no basal sliding ($\lambda = 0$) and no ice accumulation ($a = 0$).

2.5.2 Extensional ice sheets

Using the relationship

$$m = \frac{1}{n}, \qquad C = 2^{1-\frac{1}{n}}\mu_0\lambda^{-1/n}, \tag{2.138}$$

the extensional flow equations can be expressed using the same parameters as the shear-dominated model.

In the ice sheet, for $0 \leq x \leq x_G$,

$$\frac{\partial H}{\partial x} = \alpha - \frac{2^{1-\frac{1}{n}}\mu_0}{\rho g H \lambda^{1/n}}\left(\frac{q_0 + ax}{H}\right)^{1/n}, \tag{2.139a}$$

$$b = -\alpha x, \qquad h = H + b, \tag{2.139b}$$

$$u = \frac{q_0 + ax}{H} = \left[-2^{\frac{1}{n}-1}\frac{\rho g H}{\mu_0}\lambda^{1/n}\frac{\partial h}{\partial x}\right]^n. \tag{2.139c}$$

At the grounding line, $x = x_G$,

$$q_0 + ax_G = \left[2^{\frac{1}{n}-1}\frac{\rho g}{\mu_0}\lambda^{1/n}\left(\frac{\rho g'}{8\mu_0}\right)^n\right]^{\frac{n}{n+1}} H_G^{\frac{n^2+3n+1}{n+1}}, \tag{2.140a}$$

$$x_G = \frac{\rho}{\rho_w \alpha}H_G. \tag{2.140b}$$

In the ice shelf, for $x_G \leq x \leq x_N$,

$$\frac{\partial H}{\partial x} = \frac{aH - H^2 \left(\frac{\rho g'}{8\mu_0} H\right)^n}{q_0 + ax}, \tag{2.141a}$$

$$b = -\frac{\rho}{\rho_w} H, \qquad h = \left(1 - \frac{\rho}{\rho_w}\right) H, \tag{2.141b}$$

$$u = \frac{q_0 + ax}{H}. \tag{2.141c}$$

This system of governing equations has been developed previously by Schoof [21].

Apart from the evolution of the calving front $x = x_N$, these are a complete set of equations governing the ice thickness and grounding line position. As we are mainly interested in the region near the grounding line, we assume that the calving front is sufficiently far away that $x_N \to \infty$. Under this assumption, we examine steady-state solutions numerically in the following section.

Chapter 3

Steady-state numerical solutions

In this chapter, we non-dimensionalise the governing equations and solve them numerically using MATLAB. The grounding-line conditions are reduced to scalar nonlinear root-finding problems using the flotation condition and are solved using MATLAB's `fzero` function. A positive root is first bracketed, after which `fzero` uses a safeguarded combination of bisection, secant, and inverse quadratic interpolation. Once the grounding-line position and thickness have been determined, the grounded ice sheet equations are integrated upstream from $x = x_G$ to $x = 0$, while the shelf equation is integrated downstream from $x = x_G$ to a designated shelf extent. These initial-value problems are solved using MATLAB's `ode45` function, which implements an adaptive explicit Dormand–Prince Runge–Kutta $(4, 5)$ method.

3.1 Non-dimensionalisation

3.1.1 Shear-dominated ice sheets

We apply different scalings to non-dimensionalise the governing equations and boundary conditions, depending on the model and rheology, as outlined below.

Newtonian flow

Here, we non-dimensionalise Equations (2.132), (2.133), and (2.134) by introducing the scaled variables

$$H = H_0\hat{H}, \qquad h = H_0\hat{h}, \qquad b = H_0\hat{b}, \qquad x = L_0\hat{x}, \qquad N = \rho g H_0^2 \hat{N}. \tag{3.1}$$

The flux and velocity are scaled according to

$$q = q_0\hat{q}, \qquad u = \frac{q_0}{H_0}\hat{u}. \tag{3.2}$$

Following Robison et al. [17], we define the horizontal and vertical length scales by

$$H_0 = BD, \qquad L_0 = 2BD^4, \tag{3.3}$$

where

$$B = \left(\frac{6\mu q_0}{g\rho}\right)^{1/3}, \qquad D = \left(\frac{g}{g'}\right)^{1/6}. \tag{3.4}$$

Upon dropping hats, the dimensionless equations in the ice sheet, for $0 \le x \le x_G$, are

$$\frac{\partial H}{\partial x} = \beta - \frac{1 + \mathcal{A}x}{H^3 + \Lambda H^2}, \tag{3.5a}$$

$$b = -\beta x, \qquad h = H + b, \tag{3.5b}$$

$$u = -\frac{\partial h}{\partial x}\left[\Lambda H + \frac{3}{2}\left(H^2 - (h-z)^2\right)\right]. \tag{3.5c}$$

At the grounding line, $x = x_G$,

$$0 = \frac{3}{2}H_G^2 - \left(\frac{\partial h}{\partial x}\right)^2\left(\frac{3}{2}H_G^2 + \Lambda H_G\right) - \beta\Lambda\frac{\partial h}{\partial x}H_G - \mathcal{A}, \tag{3.6a}$$

$$\frac{\partial h}{\partial x} = -\frac{1 + \mathcal{A}x_G}{H_G^3 + \Lambda H_G^2}, \tag{3.6b}$$

$$H_G = (\beta + \epsilon)x_G. \tag{3.6c}$$

In the ice shelf, for $x_G \le x \le x_N$,

$$\frac{\partial H}{\partial x} = \frac{\mathcal{A}H - \frac{3}{2}H^3}{1 + \mathcal{A}x}, \tag{3.7a}$$

$$b = -\frac{\beta}{\beta + \epsilon}H, \qquad h = \frac{\epsilon}{\beta + \epsilon}H, \tag{3.7b}$$

$$u = \frac{1 + \mathcal{A}x}{H}. \tag{3.7c}$$

Here,

$$\beta = \frac{\alpha L_0}{H_0}, \qquad \mathcal{A} = \frac{aL_0}{q_0}, \qquad \Lambda = \frac{3\lambda}{H_0}, \qquad \epsilon = \beta\frac{\rho_w - \rho}{\rho}. \tag{3.8}$$

These dimensionless equations, excluding the additional effects of ice accumulation and basal sliding, have been derived previously by Robison et al. [17].

Non-Newtonian flow

We generalise the scalings defined in Equations (3.1) and (3.2) in terms of the generalised vertical and horizontal length scales

$$H_0 = B_n D_n, \qquad L_0 = \left(\frac{2^{2n+1}}{n+1}\right)^{\frac{1}{n+1}} B_n D_n^{\frac{2(n+1)}{n}}, \tag{3.9}$$

where

$$\begin{aligned} B_n &= \left[\frac{q_0(n+2)}{2^{1-n}}\left(\frac{\mu_0}{\rho g}\right)^n \left(\frac{2^{1-n}}{n+1}\right)^{\frac{n}{n+1}} 8^{\frac{n^2}{n+1}}\right]^{\frac{1}{n+2}}, \\ D_n &= \left(\frac{g}{g'}\right)^{\frac{n^2}{(n+1)(n+2)}}. \end{aligned} \tag{3.10}$$

For $n = 1$, these reduce to $H_0 = BD$ and $L_0 = 2BD^4$, recovering the scaling used in the Newtonian case. Applying these scalings to Equations (2.135), (2.136), and (2.137) yields the following system of dimensionless governing equations.

In the ice sheet, for $0 \le x \le x_G$,

$$\frac{\partial H}{\partial x} = \beta - \left[\frac{1 + \mathcal{A}x}{H^{n+2} + \Lambda H^{n+1}}\right]^{1/n}, \tag{3.11a}$$

$$b = -\beta x, \qquad h = H + b, \tag{3.11b}$$

$$u = -\frac{\partial h}{\partial x}\left|\frac{\partial h}{\partial x}\right|^{n-1}\left[\Lambda H^n + \frac{n+2}{n+1}\left(H^{n+1} - (h-z)^{n+1}\right)\right]. \tag{3.11c}$$

At the grounding line, $x = x_G$,

$$0 = \mathcal{A} + \frac{\partial h}{\partial x}\left|\frac{\partial h}{\partial x}\right|^{n-1}\left[\frac{n+2}{n+1}H_G^{n+1}\frac{\partial h}{\partial x} + \Lambda H_G^n\left(\frac{\partial h}{\partial x} + \beta\right)\right] - \frac{n+2}{n+1}H_G^{n+1}, \tag{3.12a}$$

$$\frac{\partial h}{\partial x} = -\left[\frac{1 + \mathcal{A}x_G}{H_G^{n+2} + \Lambda H_G^{n+1}}\right]^{1/n}, \tag{3.12b}$$

$$H_G = (\beta + \epsilon)x_G. \tag{3.12c}$$

In the ice shelf, for $x_G \leq x \leq x_N$,

$$\frac{\partial H}{\partial x} = \frac{\mathcal{A}H - \frac{n+2}{n+1}H^{n+2}}{1+\mathcal{A}x}, \tag{3.13a}$$

$$b = -\frac{\beta}{\beta+\epsilon}H, \qquad h = \frac{\epsilon}{\beta+\epsilon}H, \tag{3.13b}$$

$$u = \frac{1+\mathcal{A}x}{H}. \tag{3.13c}$$

Here, the bed slope parameter β, accumulation rate $\mathcal{A}$, and flotation parameter ϵ are as defined in Equation (3.8), while

$$\Lambda = (n+2)\frac{\lambda}{H_0}. \tag{3.14}$$

This system of dimensionless governing equations reduces to that of Robison et al. [17] when the flow is Newtonian ($n = 1$) and when $\Lambda = \mathcal{A} = 0$.

3.1.2 Extensional ice sheets

We non-dimensionalise the extensional flow equations (2.139), (2.140), and (2.141) by applying the same scalings (3.1), (3.2), (3.9) and (3.10) as in the shear-dominated case. This gives the following system of dimensionless equations.

In the ice sheet, for $0 \leq x \leq x_G$,

$$\frac{\partial H}{\partial x} = \beta - \left[\frac{1+\mathcal{A}x}{\Lambda H^{n+1}}\right]^{1/n}, \tag{3.15a}$$

$$b = -\beta x, \qquad h = H + b, \tag{3.15b}$$

$$u = \frac{1+\mathcal{A}x}{H}. \tag{3.15c}$$

At the grounding line, $x = x_G$,

$$1 + \mathcal{A}x_G = \left(\frac{n+2}{n+1}\right)^{\frac{n}{n+1}} \Lambda^{\frac{1}{n+1}} H_G^{\frac{n^2+3n+1}{n+1}}, \tag{3.16a}$$

$$H_G = (\beta+\epsilon)x_G. \tag{3.16b}$$

In the ice shelf, for $x_G \leq x \leq x_N$,

$$\frac{\partial H}{\partial x} = \frac{\mathcal{A}H - \frac{n+2}{n+1}H^{n+2}}{1+\mathcal{A}x}, \tag{3.17a}$$

$$b = -\frac{\beta}{\beta+\epsilon}H, \qquad h = \frac{\epsilon}{\beta+\epsilon}H, \tag{3.17b}$$

$$u = \frac{1+\mathcal{A}x}{H}. \tag{3.17c}$$

Again, the bed slope parameter β, accumulation rate $\mathcal{A}$, and flotation parameter ϵ are as defined in Equation (3.8), while slip length Λ is as defined in Equation (3.14).

This system of governing equations follows Schoof [21].

We note that the shear-dominated ice-sheet model is valid only when slip length $\Lambda \ll 1$, while the extensional ice-sheet model is valid only when slip length $\Lambda \gg 1$. We also explore both models outside these formal validity ranges to examine their asymptotic behaviour and compare the two solutions.

3.2 Exact solutions

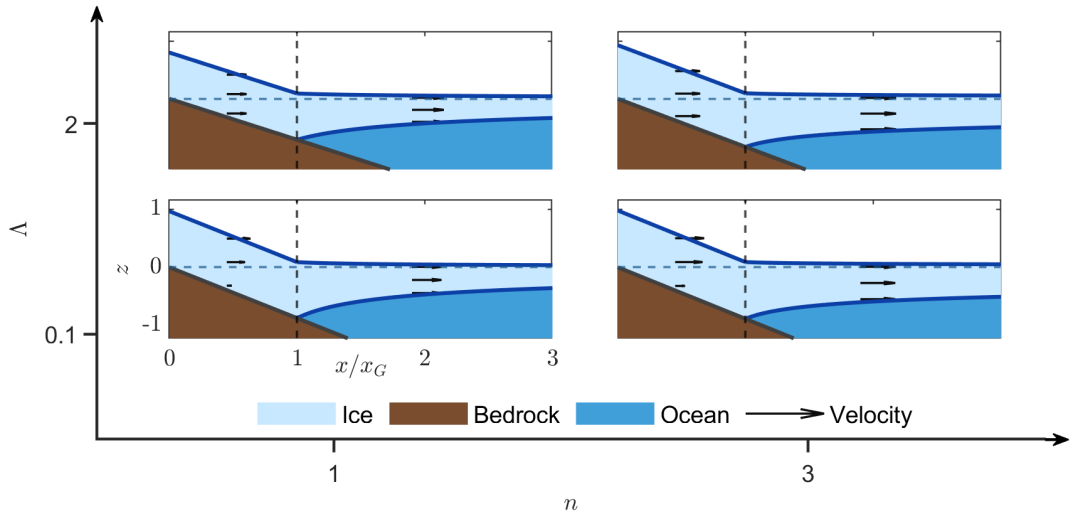


Figure 3.1: Constant-depth grounded ice-sheet profiles for $n = 1$ (left panels) and $n = 3$ (right panels), with accumulation rate $\mathcal{A} = 0$ and flotation parameter $\epsilon = 0.1$. The bottom row corresponds to shear-dominated ice sheets, valid for $\Lambda \ll 1$, where, specifically, $\Lambda = 0.1$. For $\Lambda \gg 1$, the upper row shows results using the extensional ice-sheet model, where, specifically, $\Lambda = 2$. In the shear-dominated case, the bed slope corresponding to constant sheet depth is $\beta = 1$ for both $n = 1$ and $n = 3$. In the extensional ice-sheet model, the corresponding bed slopes are $\beta = 0.7759$ for $n = 1$ and $\beta = 0.8733$ for $n = 3$.

In this section, exact solutions to the non-dimensionalised equations presented in the previous section are discussed. We note that exact solutions do not exist for typical parameter values, thus necessitating numerical solutions in general, which we obtain in later sections.

3.2.1 The ice sheet

In general, the ice-sheet equations do not have exact solutions. However, in particular idealised geometries and parameter regimes, exact solutions may be found. Some exact solutions of interest will be presented here, while we explore the remaining parameter space by solving the systems of ODEs numerically in a later section.

For both the shear-dominated and the extensional ice-sheet models, solutions of uniform ice depth, $\frac{\partial H}{\partial x} = 0$, may be found in the no-accumulation limit, $\mathcal{A} = 0$. For constant ice-sheet depth, it follows from the model geometry that $\frac{\partial h}{\partial x} = -\beta$. In the shear-dominated model, upon simplification, the grounding-line stress condition becomes

$$0 = \frac{n+2}{n+1} H_G^{n+1} \left(1 - \beta^{n+1}\right). \tag{3.18}$$

Thus, for no accumulation, constant depth occurs for bed slope $\beta = 1$. Note that this has no dependence on slip length Λ or rheological exponent n. The sheet equation is therefore given by

$$H_G^{n+1}(H_G + \Lambda) = 1 \tag{3.19}$$

This is an extension of the Newtonian, no-slip constant-depth solution presented in Robison et al. [17] to account for shear-thinning rheology and basal slip.

In the extensional ice-sheet model, the grounding-line flux for no accumulation returns

$$H_G = \left[\left(\frac{n+1}{n+2}\right)^n \Lambda^{-1}\right]^{1/(n^2+3n+1)}, \tag{3.20}$$

which can be combined with the constant-depth sheet solution to give

$$\beta = \left[\frac{1}{\Lambda H_G^{n+1}}\right]^{1/n}, \tag{3.21}$$

thus determining the slope at which the constant-depth profile occurs. It should be noted that in the extensional ice-sheet model, the slope required for constant depth is dependent on the slip length Λ and rheological exponent n, whereas the shear-dominated ice-sheet model is independent of Λ and n. Figure 3.1 presents the constant-depth profiles for

different values of the slip length Λ and rheological exponent n, agreeing with the independence of the shear-dominated ice-sheet model while demonstrating the n-dependence of the extensional ice-sheet model.

3.2.2 The grounding line

Again, in the general case, the grounding-line equations do not have an explicit solution but solutions do exist for particular idealised geometries and parameter regimes.

In the no-accumulation limit, the grounding-line position may be found explicitly in the extensional ice-sheet model as

$$x_G = \frac{1}{\beta+\epsilon}\left[\left(\frac{n+1}{n+2}\right)^n \frac{1}{\Lambda}\right]^{\frac{1}{n^2+3n+1}}. \tag{3.22}$$

In the no-slip, zero-accumulation limit, the free-surface slope of the shear-dominated ice sheet at the grounding line becomes

$$\frac{\partial h}{\partial x} = -H_G^{-\frac{n+2}{n}}. \tag{3.23}$$

In this limit, the grounding-line stress condition reduces to

$$0 = \frac{n+2}{n+1} H_G^{n+1}\left[H_G^{-\frac{(n+1)(n+2)}{n}} - 1\right], \tag{3.24}$$

from which

$$H_G = 1, \qquad x_G = \frac{1}{\beta+\epsilon}. \tag{3.25}$$

This generalises the Newtonian result obtained by Robison et al. [17] to non-Newtonian flow. Furthermore, for the constant-depth solution, for which $\beta = 1$, we have $H = H_G = 1$ throughout the ice sheet. Since $h = H - \beta x$, it follows that

$$h = 1 - x, \tag{3.26}$$

so that the linear sheet profile identified by Robison et al. [17] in the Newtonian limit also holds here for non-Newtonian flow.

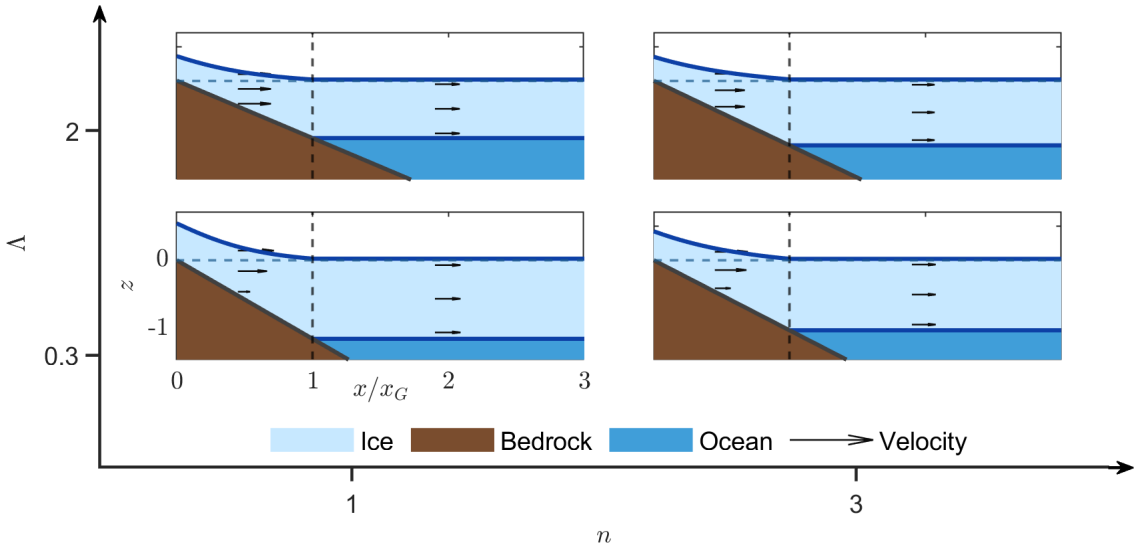


Figure 3.2: Constant-shelf-depth profiles for $n = 1$ (left panels) and $n = 3$ (right panels), with bed slope $\beta = 5$ and flotation parameter $\epsilon = 0.1$. The bottom row corresponds to shear-dominated ice sheets, valid for $\Lambda \ll 1$, where, specifically, $\Lambda = 0.1$. For $\Lambda \gg 1$, the upper row shows results using the extensional ice-sheet model, where, specifically, $\Lambda = 2$. In the shear-dominated case, the accumulation rate corresponding to constant shelf depth is $\mathcal{A} = 2.06$ and $\mathcal{A} = 1.488$ for $n = 1$ and $n = 3$, respectively. In the extensional ice-sheet model, the corresponding accumulation rates are $\mathcal{A} = 1.109$ for $n = 1$ and $\mathcal{A} = 1.101$ for $n = 3$.

3.2.3 The ice shelf

In contrast to the ice sheet, the ice shelf does have explicit solutions in general.

In the no-accumulation limit, the shelf equation reduces to

$$\frac{\partial H}{\partial x} = -\frac{n+2}{n+1} H^{n+2}, \tag{3.27}$$

which is a separable equation with a solution

$$H(x) = \left[H_G^{-(n+1)} + (n+2)(x - x_G) \right]^{-\frac{1}{n+1}}. \tag{3.28}$$

Thus, in the no-accumulation limit, $H(x) \to 0$ as $x \to \infty$.

In the case of non-zero accumulation, a solution to the shelf equation can be found by defining $Y = H^{-(n+1)}$, which returns the linear ODE

$$\frac{\partial Y}{\partial x} + \frac{(n+1)\mathcal{A}}{1+\mathcal{A}x} Y = \frac{n+2}{1+\mathcal{A}x}. \tag{3.29}$$

This has the solution

$$H(x) = \left[\frac{n+2}{(n+1)\mathcal{A}} + \left(\frac{1+\mathcal{A}x_G}{1+\mathcal{A}x} \right)^{n+1} \left(H_G^{-(n+1)} - \frac{n+2}{(n+1)\mathcal{A}} \right) \right]^{-\frac{1}{n+1}}, \tag{3.30}$$

which reduces to Equation (3.28) when $\mathcal{A} = 0$. Note that

$$H \to \left[\frac{n+1}{n+2} \mathcal{A} \right]^{\frac{1}{n+1}} \neq 0 \tag{3.31}$$

as $x \to \infty$. Thus, for large accumulation, the shelf depth may increase.

The constant shelf depth may be found by equating the grounding-line depth to the asymptotic shelf depth, returning

$$\mathcal{A} = \frac{n+2}{n+1} H_G^{n+1}. \tag{3.32}$$

It is notable that for this value of accumulation, the shelf depth equation reduces to $\frac{\partial H}{\partial x} = 0$. The grounding-line depth, H_G, can then be determined from the grounding-line conditions. In the shear-dominated ice-sheet model, the stress condition reduces to

$$0 = \frac{n+2}{n+1} H_G^{n+1} \frac{\partial h}{\partial x} + \Lambda H_G^n \left(\frac{\partial h}{\partial x} + \beta \right) \qquad \text{at} \qquad x = x_G, \tag{3.33}$$

where the contribution $\frac{n+2}{n+1} H_G^{n+1}$ associated with longitudinal stretching within the shelf is balanced by the effect of accumulation. When slip length $\Lambda = 0$, the only solution is $H_G = 0$, and hence no solution with positive constant shelf depth exists in the no-slip limit.

For the extensional ice-sheet model, the grounding-line boundary conditions may be solved together with the accumulation condition given by Equation (3.32). This determines H_G, and consequently the corresponding value of accumulation rate $\mathcal{A}$.

Figure 3.2 shows the resulting constant-shelf profiles for both the Newtonian and non-Newtonian flow. For accumulation rates greater than the values indicated in the figure, the shelf depth increases towards its asymptotic limit, whereas for smaller accumulation rates, it decreases towards the same limit.

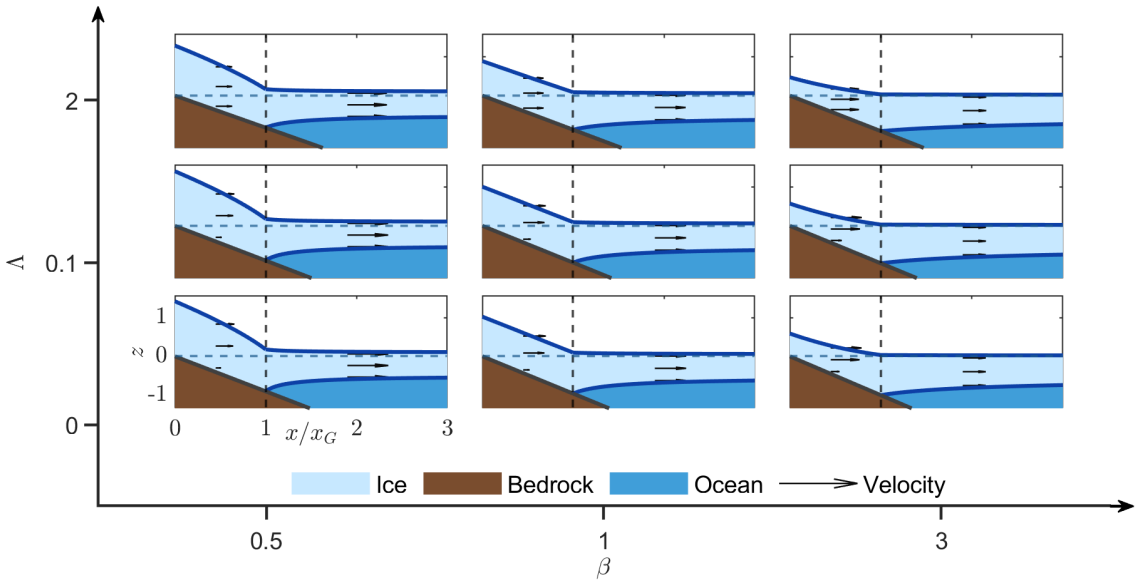


Figure 3.3: Ice-sheet and ice-shelf profiles for $n = 3$, with accumulation rate $\mathcal{A} = 0.2$ and flotation parameter $\epsilon = 0.1$. The bottom and middle rows correspond to shear-dominated ice sheets, valid for $\Lambda \ll 1$. The bottom row shows ice sheets under no slip with $\Lambda = 0$, while the middle row shows partially lubricated ice sheets with $\Lambda = 0.1$. For $\Lambda \gg 1$, the top row shows results using the extensional ice-sheet model, for which $\Lambda = 2$. The left-hand, middle and right-hand columns correspond to bed slopes $\beta = 0.5, 1$, and 3, respectively.

3.3 Full numerical solutions

In this section, the effects of varying several key non-dimensional parameters are examined for both the shear-dominated and extensional ice-sheet models by solving the full system of governing differential equations numerically. The resulting velocity fields and thickness profiles, generated using MATLAB, are compared and interpreted physically.

We note again that the shear-dominated model is valid only when the basal slip is moderate ($\Lambda \ll 1$) while the extensional model is valid only when the basal slip is significant ($\Lambda \gg 1$).

3.3.1 Variations in bed slope and slip length

Figure 3.3 shows the effects of varying the bed slope β for the shear-dominated ice-sheet model with no-slip, ($\Lambda = 0$) and with partial-slip ($\Lambda = 0.1$) in the bottom and middle panels respectively. As the shear-dominated model is valid only for $\Lambda \ll 1$, we instead present a typical numerical solution using the extensional ice-sheet model when $\Lambda = 2$

in the upper panel of Figure 3.3, which is valid when $\Lambda \gg 1$. We chose to present the numerical results for non-Newtonian flow with $n = 3$, which is geophysically relevant for modelling ice.

The flow profiles for both the no-slip and partial-slip shear-dominated ice-sheet models and the extensional ice-sheet model show results similar to those described in Robison et al. [17], where the free upper surface is approximately parallel to the bed slope near $x = 0$, but for $\beta < 1$, the free upper-surface slope becomes more negative as the sheet approaches the grounding line and for bed slope $\beta > 1$, the free upper-surface slope becomes less negative as the sheet approaches the grounding line. For bed slope $\beta = 1$, the upper surface of the sheet is approximately parallel to the bed slope for its entire extent. This agrees with the constant-sheet-depth derivation in the previous section, where it was found that in the no-accumulation shear-dominated ice-sheet model, bed slope $\beta = 1$ described constant sheet depth for no-accumulation flow. For the extensional ice-sheet model, the bed slope value corresponding to constant sheet depth for the parameter values used in Figure 3.3 was given in the previous section as $\beta \approx 0.8733$. It is therefore unsurprising that the sheet depth looks constant for bed slope $\beta = 1$ in the low-accumulation rate $\mathcal{A} = 0.2$ profiles given in Figure 3.3.

From Equations (3.11) and (3.15), it follows that bed slopes shallower than the value required for constant ice thickness give $\frac{\partial H}{\partial x} < 0$, corresponding to a decreasing ice thickness and, therefore, a free upper-surface slope that becomes more negative as it approaches the grounding line. Conversely, steeper bed slopes give $\frac{\partial H}{\partial x} > 0$, corresponding to an increasing ice thickness and, therefore, a free upper-surface slope that becomes less negative as it approaches the grounding line. It should be noted that, in the case in which accumulation is non-zero, shown in Figure 3.3, the sheet thickness is governed by the full Equations (3.11) and (3.15). Consequently, imposing $\frac{\partial H}{\partial x} = 0$ does not yield a solution that remains valid for all $x \in [0, x_G]$ in either the shear-dominated or extensional ice-sheet model.

It is also evident from Figure 3.3 that the grounding-line depth decreases as Λ increases. This is expected on physical grounds as well lubricated ice sheets flow faster and thus become thinner. It should be noted that thinning is only modest for the parameter values displayed in Figure 3.3, in which $\Lambda = O(1)$, but the greater variety of parameter values explored in the parameter sweeps in Section 3.4 confirms that the ice sheet thins as Λ increases more significantly.

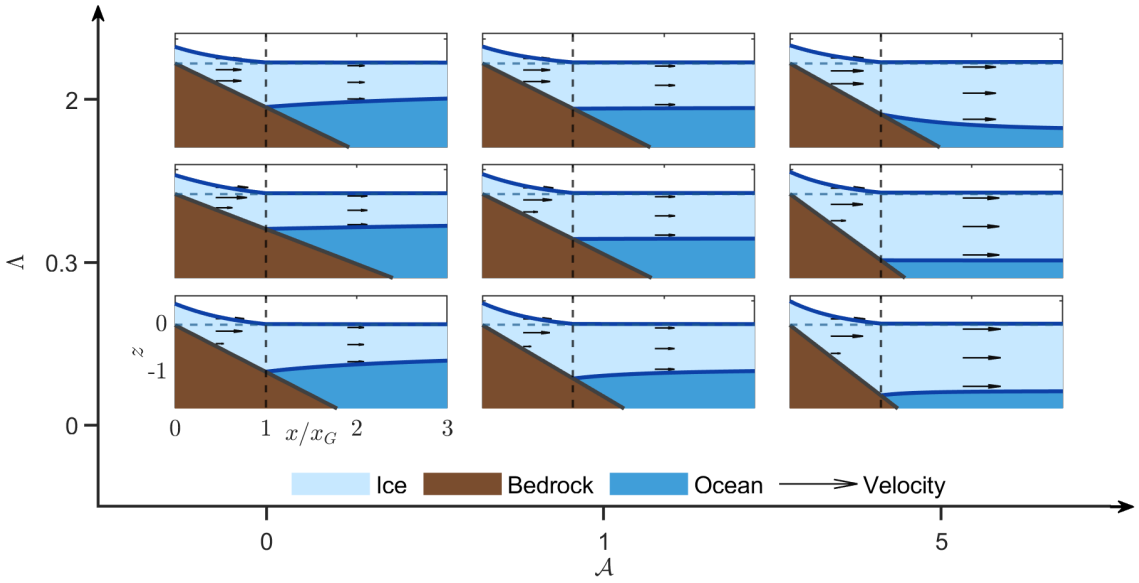


Figure 3.4: Ice-sheet and ice-shelf profiles for $n = 3$, with bed slope $\beta = 5$ and flotation parameter $\epsilon = 0.1$. The bottom and middle rows correspond to shear-dominated ice sheets, valid for $\Lambda \ll 1$. The bottom row shows ice sheets under no slip with $\Lambda = 0$, while the middle row shows partially lubricated ice sheets with $\Lambda = 0.3$. For $\Lambda \gg 1$, the top row shows results from the extensional ice-sheet model, for which $\Lambda = 2$. The left-hand, middle and right-hand columns correspond to accumulation rates $\mathcal{A} = 0, 1$ and 5, respectively.

3.3.2 Variation in accumulation rate and slip length

Figure 3.4 shows the effect of increasing the non-dimensional accumulation rate, $\mathcal{A}$, on the ice-stream profiles for the no-slip and partial-slip shear-dominated ice-sheet models, as well as the extensional ice-sheet model, for non-Newtonian flow with rheological exponent $n = 3$.

The most pronounced differences are observed in the shape of the ice shelf. In general, the shelf depth increases with accumulation rate $\mathcal{A}$ and, for sufficiently large values of accumulation rate $\mathcal{A}$, also increases downstream in both the shear-dominated and extensional ice-sheet models. This is consistent with the analysis of the previous section, where the shelf depth was shown to approach $\left[\frac{n+1}{n+2}\mathcal{A}\right]^{\frac{1}{n+1}}$ for large x. If this asymptotic thickness exceeds the grounding-line thickness, the shelf thickens downstream; if it is smaller, the shelf thins. The minimum values of accumulation rate $\mathcal{A}$ required for downstream thickening, using the parameter values from Figure 3.4, are shown in Figure 3.2. The behaviour of the profiles is consistent with these critical values. The shelf thins in the no-slip case, in

the shear-dominated ice-sheet models with no accumulation and with accumulation rate $\mathcal{A} = 1$, and in the extensional ice-sheet model with no accumulation. For the extensional ice-sheet model with accumulation rate $\mathcal{A} = 1$, the shelf depth remains approximately constant, consistent with the critical accumulation rate value $\mathcal{A} \approx 1.1$ shown in Figure 3.2 for non-Newtonian flow with rheological exponent $n = 3$. By contrast, when $\mathcal{A} = 5$, the shelf thickens in both the partial slip and extensional ice-sheet models, since this value lies above the corresponding critical accumulation rate in both cases. It is notable that, even at this large accumulation rate, the no-slip model exhibits a decreasing shelf thickness, consistent with the theoretical result that no positive solution for a constant shelf depth exists for the no-slip, shear-dominated ice-sheet model.

Note also that in the no-accumulation limit, the shear-dominated ice-sheet grounding-line depth H_G is equal to one and $x_G = \frac{1}{\beta+\epsilon}$, as predicted in the previous section. Also, the grounding-line position for the extensional ice sheet within this limit agrees with the explicit solution given in Equation (3.22), showing agreement between the numerical and analytical models.

Note that Figure 3.3 shows that the grounding-line depth decreases as Λ increases. This again confirms our hypothesis that, owing to the faster speeds within the ice sheet from increased lubrication, the ice sheet becomes thinner, the effects of which are explored further in Section 3.4.

3.4 Grounding line dynamics

We compute the grounding-line position while varying each parameter over a range of values, with the slip length Λ spanning 10^{-4} to 10^{6}. It was established in previous sections that the shear-dominated ice-sheet model is valid for small slip lengths ($\Lambda \ll 1$). However, it is also of interest to examine the behaviour of the model outside this limit and to determine whether there are values of Λ for which the shear-dominated and extensional ice-sheet models agree. When investigating the power-law dependence on slip length Λ, we also explore the effect of varying the bed slope β.

We illustrate in Figures 3.5–3.7 various power laws that emerge as the parameters of interest are either small or large. The power laws for the shear-dominated ice-sheet model are represented by the black triangles in each subplot and the power laws for the extensional

ice sheet model are represented by the red, dashed triangles.

The power-law relationships are derived mathematically and interpreted geophysically below.

3.4.1 Asymptotic solutions for large slip

As seen in panel (a) of Figures 3.5–3.7, there appears to be a clear power-law relationship between slip length Λ and grounding-line position x_G, given by $x_G \propto \Lambda^{-\frac{1}{n+1}}$ and $x_G \propto \Lambda^{-\frac{1}{n^2+3n+1}}$ for $\Lambda \gg 1$ for the shear-dominated and extensional ice-sheet models, respectively, which we derive analytically below. In general, the two models do not agree on the grounding line position. However, we can see the best agreement for moderate values of Λ, which is expected as the two models are valid for either $\Lambda \ll 1$ or $\Lambda \gg 1$. The governing equations provide justification for the power-law relationships, which we present below.

The power-law relationship in the shear-dominated case can be determined from Equations (3.12). Considering $\Lambda \gg H_G$, the free upper-surface slope at the grounding line is given by

$$\frac{\partial h}{\partial x} = -\left(\frac{1+\mathcal{A}x_G}{\Lambda H_G^{n+1}}\right)^{1/n}. \tag{3.34}$$

Combined with the stress condition, we obtain the expression

$$0 = \mathcal{A} + \frac{n+2}{n+1}\frac{(1+\mathcal{A}x_G)\left(-\frac{\partial h}{\partial x}\right)}{H_G+\Lambda} - \frac{\Lambda(1+\mathcal{A}x_G)}{H_G(H_G+\Lambda)}\left(\beta + \frac{\partial h}{\partial x}\right) - \frac{n+2}{n+1}H_G^{n+1}. \tag{3.35}$$

Applying $\Lambda \gg H_G$, the equation simplifies to

$$0 = \mathcal{A} - \frac{1+\mathcal{A}x_G}{H_G}\left(\beta + \frac{\partial h}{\partial x}\right) - \frac{n+2}{n+1}H_G^{n+1}. \tag{3.36}$$

Upon rearranging, the expression becomes

$$\beta + \frac{\partial h}{\partial x} = \frac{H_G}{1+\mathcal{A}x_G}\left[\mathcal{A} - \frac{n+2}{n+1}H_G^{n+1}\right]. \tag{3.37}$$

Ice sheets that move over well-lubricated surfaces flow faster and so are necessarily thinner. Therefore, the grounding line depth $H_G \to 0$ as $\Lambda \to \infty$ and we must have $\beta \sim -\frac{\partial h}{\partial x}$. Upon application of these scalings, Equation (3.34) becomes

$$H_G^{n+1} \sim \frac{1+\mathcal{A}x_G}{\beta^n \Lambda}. \tag{3.38}$$

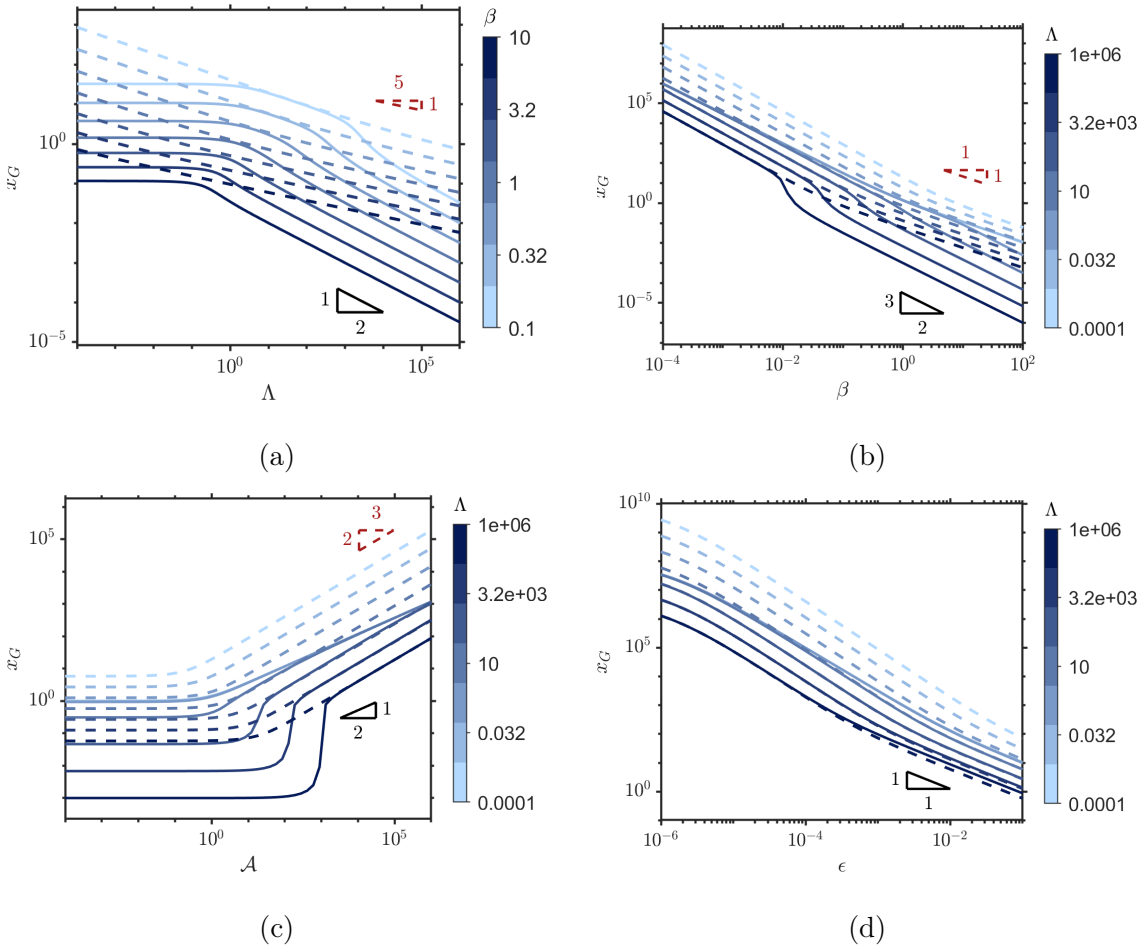


Figure 3.5: Grounding line position x_G for Newtonian flow $n = 1$, comparing shear-dominated (solid curves) and extensional ice sheet (dotted curves) models when (a) the slip length Λ varies while $\mathcal{A} = 1$, $\epsilon = 10^{-6}$ and $\beta \in [10^{-1}, 10]$; (b) the bed slope β varies while $\Lambda \in \left[10^{-4}, 10^{6}\right]$, $\mathcal{A} = 1$ and $\epsilon = 10^{-6}$; (c) the accumulation rate $\mathcal{A}$ varies while $\Lambda \in \left[10^{-4}, 10^{6}\right]$, $\beta = 1$ and $\epsilon = 10^{-6}$; and (d) the flotation parameter ϵ varies while $\Lambda \in \left[10^{-4}, 10^{6}\right]$, $\mathcal{A} = 10^{-2}$ and $\beta = 10^{-6}$. The power-law relationships are represented by the black, solid triangle and the red, dashed triangle for the shear-dominated and extensional ice sheet models, respectively.

From the flotation condition, for fixed bed slope β and flotation parameter ϵ, $x_G \propto H_G$ and so $x_G \to 0$ as $\Lambda \to \infty$, giving $\mathcal{A}x_G \ll 1$ in well-lubricated sheets. Hence, $H_G \propto \Lambda^{-1/(n+1)}$ and, applying the flotation condition, we arrive at the power-law relationship

$$x_G \propto \Lambda^{-1/(n+1)}, \tag{3.39}$$

for slip length $\Lambda \gg 1$ for the shear-dominated model, which is in agreement with the power law shown in panel (a) of Figures 3.5–3.7. Strictly speaking, this limit is outside of the domain of validity ($\Lambda \ll 1$) of the shear-dominated model; however, we find it useful to determine this power law anyway for comparison with the extensional model, which we determine below. The dependence of the bed slope β in Equation (3.38) explains the vertical separation of the curves in the grounding-line position in the top-left panels of Figures 3.5–3.7 and also explains why, for small values of β, larger values of Λ are needed for the power-law to be apparent.

We determine a power law for the grounding line position versus the slip length in the $\Lambda \gg 1$ limit for the extensional sheet model from Equations (3.16). Again making the assumption that for $\Lambda \gg 1$, grounding-line depth $H_G \ll 1$ and therefore $\mathcal{A}x_G \ll 1$, the flux condition simplifies to

$$H_G = \left(\frac{n+1}{n+2}\right)^{\frac{n}{n^2+3n+1}} \Lambda^{-\frac{1}{n^2+3n+1}}. \tag{3.40}$$

From this, after applying the flotation condition, it simply follows that for fixed bed slope β and flotation parameter ϵ,

$$x_G \propto \Lambda^{-\frac{1}{n^2+3n+1}}, \tag{3.41}$$

for $\Lambda \gg 1$, which is in agreement with the power law shown in panel (a) of Figures 3.5–3.7. It should be noted that $n^2 + 3n + 1 > n + 1$ for $n > 0$, explaining the shallower gradient of the power law in the extensional ice-sheet model, particularly at higher values of rheological exponent n.

3.4.2 Asymptotic solutions for small and large bed slope

As seen in panel (b) of Figures 3.5–3.7, there appears to be a clear power-law relationship between bed slope β and grounding-line position x_G, given by $x_G \propto \beta^{-\frac{2n+1}{n+1}}$ and $x_G \propto \beta^{-1}$ for $\beta \gg 1$ for the shear-dominated and extensional ice-sheet models, respectively, which we derive analytically below. The two models appear to agree on grounding-line position

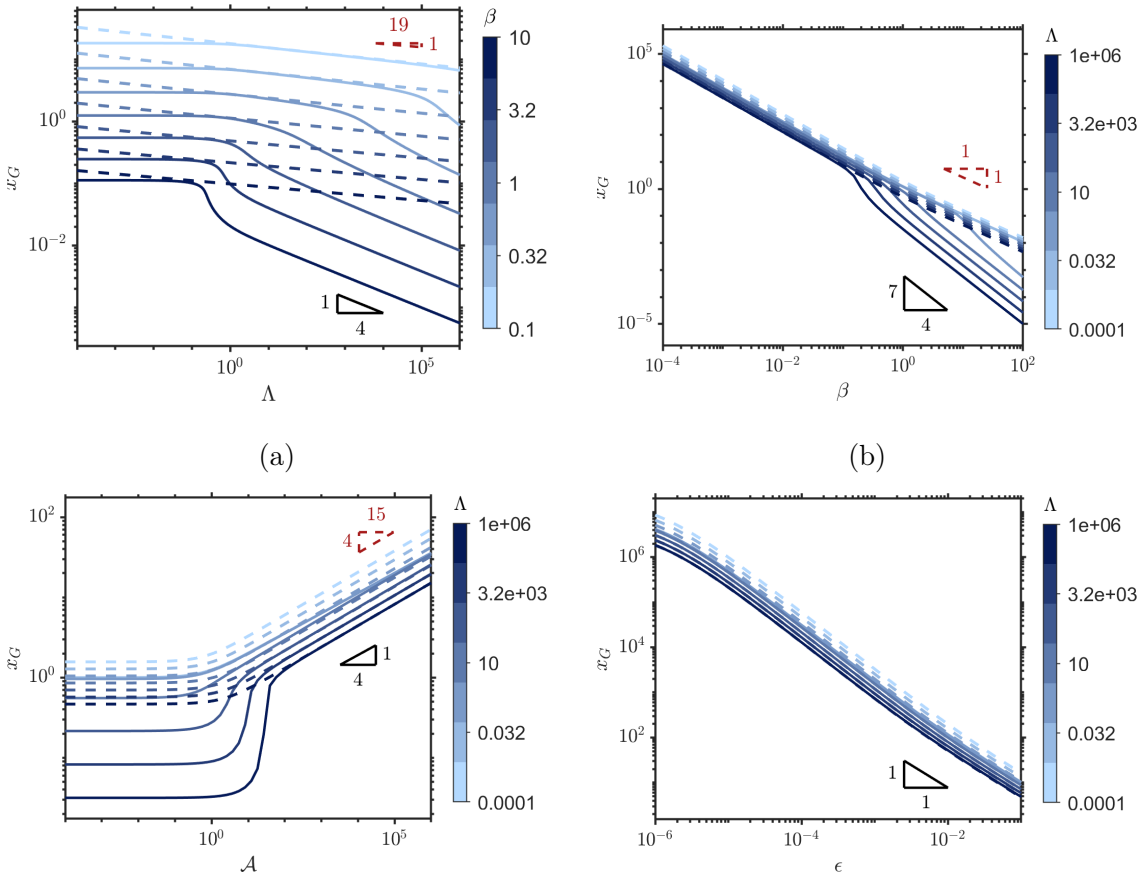


Figure 3.6: Grounding line position x_G for non-Newtonian flow $n = 3$, comparing shear-dominated (solid curves) and extensional ice sheet (dotted curves) models when (a) the slip length Λ varies while $\mathcal{A} = 1$, $\epsilon = 10^{-6}$ and $\beta \in [10^{-1}, 10]$; (b) the bed slope β varies while $\Lambda \in \left[10^{-4}, 10^{6}\right]$, $\mathcal{A} = 1$ and $\epsilon = 10^{-6}$; (c) the accumulation rate $\mathcal{A}$ varies while $\Lambda \in \left[10^{-4}, 10^{6}\right]$, $\beta = 1$ and $\epsilon = 10^{-6}$; and (d) the flotation parameter ϵ varies while $\Lambda \in \left[10^{-4}, 10^{6}\right]$, $\mathcal{A} = 10^{-2}$ and $\beta = 10^{-6}$. The power-law relationships are represented by the black, solid triangle and the red, dashed triangle for the shear-dominated and extensional ice sheet models, respectively.

for shallow bed slopes and large slip lengths. However, for steeper bed slopes, the models follow quite different power laws. The governing equations provide justification for these power-law relationships for both the shear-dominated and extensional ice sheet models, which we present below.

The power law for the extensional model is obtained by considering Equations (3.16). For steep bed slopes, the depth needed for flotation to occur due to Archimedes' principle is obtained for smaller values of x, and, therefore, $x_G \to 0$ as $\beta \to \infty$. Therefore, since $\mathcal{A}x_G \ll 1$, the flux condition reduces to Equation (3.40) and so grounding-line depth H_G is independent of β. Thus, the bed slope β only affects the grounding-line position through the flotation condition $x_G = \frac{H_G}{\beta+\epsilon}$. Therefore, for $\beta \gg \epsilon$, the power law is simply

$$x_G \propto \beta^{-1} \tag{3.42}$$

for $\beta \gg 1$ which is in agreement with the power law shown in panel (b) of Figures 3.5–3.7.

To determine the power-law relationship between bed slope and grounding line position for the shear-dominated ice-sheet model, we consider the set of Equations (3.12). We note that for a steeper bed slope, gravity drives faster basal sliding and greater deformation within the ice sheet. This results in faster velocities within the ice sheet, causing it to thin. Therefore for large values of β, grounding-line depth H_G is small so that $H_G \ll \Lambda$. The free upper-surface slope at the grounding line therefore reduces to Equation (3.34) and the stress condition to Equation (3.37).

So again, we have the relationship described in Equation (3.38), yielding the power-law relationship

$$H_G \propto \beta^{-\frac{n}{n+1}}. \tag{3.43}$$

From the flotation condition, $x_G \propto H_G\beta^{-1}$. Therefore, we obtain the power-law relationship

$$x_G \propto \beta^{-\frac{2n+1}{n+1}} \tag{3.44}$$

for $\beta \gg 1$, which is in agreement with the power law shown in panel (b) of Figures 3.5–3.7.

The slip-length dependence in Equations (3.38) and (3.40) explains the vertical spread for different Λ values. This also explains why, for small values of slip length, a steeper bed slope is required for the power laws to be apparent.

The agreement at small bed slope β and large slip length Λ between the two models follows from the stress condition (3.37) for the shear-dominated ice-sheet model. Using the inverse

of the geophysical reasoning provided earlier, shallow bed slopes reduce basal sliding and deformation due to gravity, resulting in slower velocities and therefore thicker ice sheets. Thus $H_G \gg 1$ for $\beta \ll 1$. Within these limits, upon simplification, the stress condition becomes

$$\frac{\partial h}{\partial x} = -\frac{n+2}{n+1}\frac{H_G^{n+2}}{1+\mathcal{A}x_G}. \tag{3.45}$$

Comparing with the free upper-surface slope at the grounding line for large slip length Λ determined by Equation (3.34) and upon rearranging we obtain the relationship

$$1+\mathcal{A}x_G = \left(\frac{n+2}{n+1}\right)^{\frac{n}{n+1}} \Lambda^{\frac{1}{n+1}} H_G^{\frac{n^2+3n+1}{n+1}}. \tag{3.46}$$

This is exactly the flux condition at the grounding line used in the extensional ice-sheet model, so for large slip length Λ, and a shallow bed slope, the two models agree on the grounding-line position. For small slip length Λ, this assumption breaks down and so we have quite different grounding-line positions between the two models.

3.4.3 Asymptotic solutions for large accumulation

As seen in panel (c) of Figures 3.5–3.7, there appears to be a clear power-law relationship between the accumulation $\mathcal{A}$ and the grounding-line position x_G. For the shear-dominated ice-sheet model, $x_G \propto \mathcal{A}^{\frac{1}{n+1}}$ for $H_G \gg \Lambda$, while $x_G \propto \mathcal{A}^{\frac{n+1}{n(n+2)}}$ for $\Lambda \gg H_G$. The latter power law also applies to the extensional ice-sheet model. For large values of accumulation, the two models agree very closely; however, there are large differences in the grounding-line positions at small values of accumulation. We provide justification for these claims, based on the governing equations and geophysical reasoning, below.

The power-law relationship for the shear-dominated ice-sheet model can be determined by noting that for large rates of accumulation, the ice sheet thickens and so greater water depth is needed for flotation to occur. Therefore, the grounding-line position moves significantly downstream, and so $\mathcal{A}x_G \gg 1$ and $1+\mathcal{A}x_G \sim \mathcal{A}x_G$. As grounding-line position x_G grows, from the flotation condition, so does grounding-line depth H_G and so for large enough accumulation, we have $H_G \gg \Lambda$. Applying the flotation condition, the slope at the grounding line may be written as

$$\frac{\partial h}{\partial x} = -\left(\frac{\mathcal{A}}{(\beta+\epsilon)H_G^{n+1}}\right)^{1/n}. \tag{3.47}$$

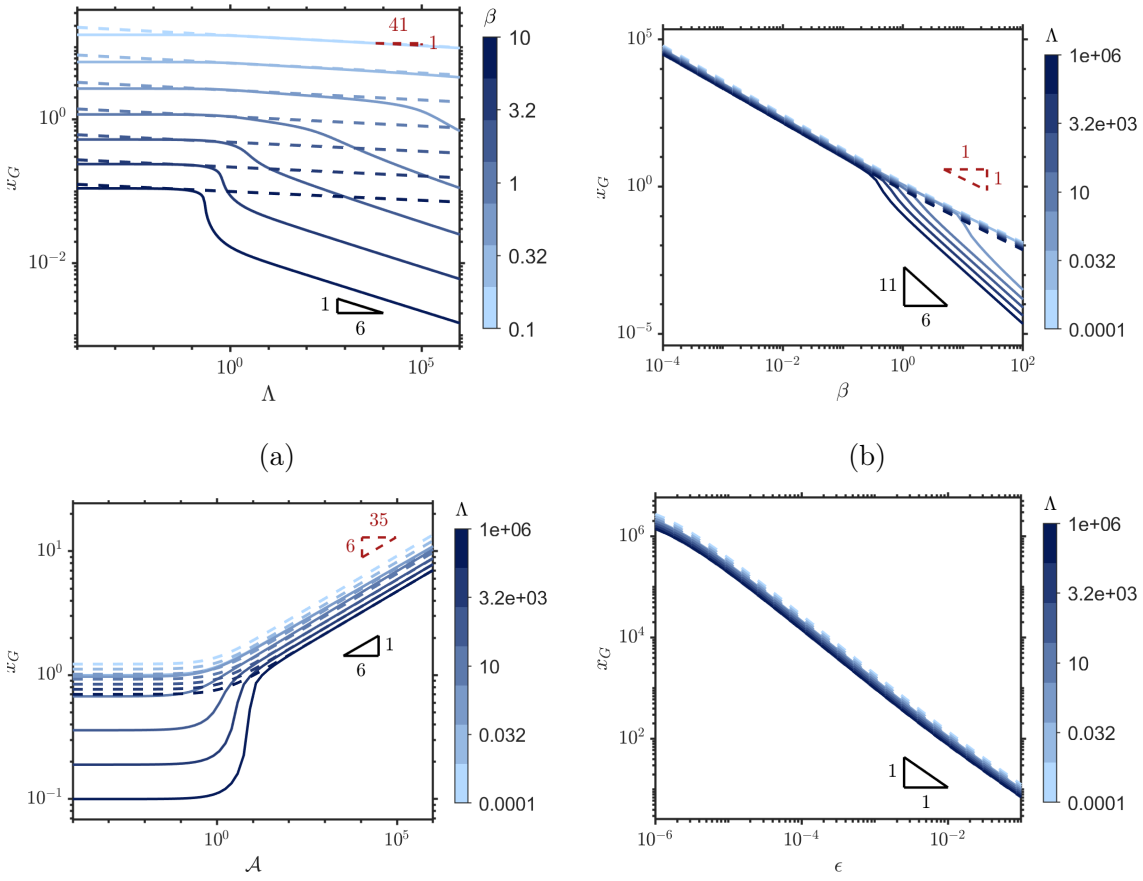


Figure 3.7: Grounding line position x_G for non-Newtonian flow $n = 5$, comparing shear-dominated (solid curves) and extensional ice sheet (dotted curves) models when (a) the slip length Λ varies while $\mathcal{A} = 1$, $\epsilon = 10^{-6}$ and $\beta \in [10^{-1}, 10]$; (b) the bed slope β varies while $\Lambda \in \left[10^{-4}, 10^{6}\right]$, $\mathcal{A} = 1$ and $\epsilon = 10^{-6}$; (c) the accumulation rate $\mathcal{A}$ varies while $\Lambda \in \left[10^{-4}, 10^{6}\right]$, $\beta = 1$ and $\epsilon = 10^{-6}$; and (d) the flotation parameter ϵ varies while $\Lambda \in \left[10^{-4}, 10^{6}\right]$, $\mathcal{A} = 10^{-2}$ and $\beta = 10^{-6}$. The power-law relationships are represented by the black, solid triangle and the red, dashed triangle for the shear-dominated and extensional ice sheet models, respectively.

For slip lengths $\Lambda \ll H_G$, the grounding line stress condition may be written as

$$0 = \mathcal{A} + \frac{n+2}{n+1} H_G^{n+1} \left(-\frac{\partial h}{\partial x} \right)^{n+1} - \frac{n+2}{n+1} H_G^{n+1}. \tag{3.48}$$

Dividing by H_G^{n+1} and applying the expression for $\frac{\mathcal{A}}{H_G^{n+1}}$ obtained by rearranging Equation (3.47), we obtain the expression

$$(\beta + \epsilon) \left(-\frac{\partial h}{\partial x} \right)^{n} + \frac{n+2}{n+1} \left(-\frac{\partial h}{\partial x} \right)^{n+1} = \frac{n+2}{n+1} \tag{3.49}$$

Therefore, $\frac{\partial h}{\partial x}$ is independent of the accumulation rate $\mathcal{A}$ for large rates of accumulation and is approximately $O(1)$. Thus, from Equation (3.47), $\mathcal{A} \propto H_G^{n+1}$ and therefore, by flotation, we obtain the power-law relation

$$x_G \propto \mathcal{A}^{\frac{1}{n+1}} \tag{3.50}$$

for $H_G \gg \Lambda$, which is in agreement with the power law shown in panel (c) of Figures 3.5–3.7.

Note that this power law only holds in the limit $H_G \gg \Lambda$. In the plots, we vary the slip length across 10 orders of magnitude, specifically, $\Lambda \in [10^{-4}, 10^{6}]$, and the grounding-line position x_G does not exceed $O(10)$. Therefore, the case $\Lambda \gg H_G$ should also be considered. The grounding line upper free-surface slope in this limit is given by

$$\frac{\partial h}{\partial x} = -\left(\frac{\mathcal{A} x_G}{\Lambda H_G^{n+1}} \right)^{1/n}. \tag{3.51}$$

The stress condition for large slip length Λ and accumulation rate $\mathcal{A}$ reduces to

$$0 = \mathcal{A} - \frac{\mathcal{A} x_G}{H_G} \left(\beta + \frac{\partial h}{\partial x} \right) - \frac{n+2}{n+1} H_G^{n+1}. \tag{3.52}$$

Rearranging and applying the flotation condition, we obtain the expression

$$\mathcal{A} \frac{\epsilon - \frac{\partial h}{\partial x}}{\beta + \epsilon} = \frac{n+2}{n+1} H_G^{n+1}. \tag{3.53}$$

Under the assumption that the flotation parameter ϵ is small compared to the bed slope β and the free upper-surface slope $\frac{\partial h}{\partial x}$, we have

$$\frac{\partial h}{\partial x} = -\frac{n+2}{n+1} \frac{\beta}{\mathcal{A}} H_G^{n+1}. \tag{3.54}$$

Therefore, we obtain the relationship

$$\left(-\frac{\partial h}{\partial x} \right)^{n} \propto \frac{H_G^{n(n+1)}}{\mathcal{A}^n}. \tag{3.55}$$

From Equation (3.51), we have the power law

$$\left(-\frac{\partial h}{\partial x}\right)^n \propto \frac{\mathcal{A}}{H_G^n}. \tag{3.56}$$

Comparing the two relationships and applying the flotation condition, we arrive at the power-law relationship

$$x_G \propto \mathcal{A}^{\frac{n+1}{n(n+2)}}, \tag{3.57}$$

for $\Lambda \gg H_G$, which is in agreement with the power law shown in panel (c) of Figures 3.5–3.7. Note that $\frac{n+1}{n(n+2)} - \frac{1}{n+1} = \frac{1}{n(n+1)(n+2)} \sim \frac{1}{n^3}$; therefore, for large values of the rheological exponent n, the power laws are approximately identical. Also note that this power law only holds in the case that $\Lambda \gg H_G$, which occurs for the finite value of accumulation rate considered here. In the limit $\mathcal{A} \to \infty$, we have $H_G \gg \Lambda$, and, therefore, the power law in Equation (3.50) holds.

For the extensional sheet model, again using the assumption that for large accumulation, $1 + \mathcal{A}x_G \sim \mathcal{A}x_G$, the grounding-line flux condition, upon enforcing flotation, becomes

$$\mathcal{A}x_G = \left(\frac{n+2}{n+1}\right)^{\frac{n}{n+1}} \Lambda^{\frac{1}{n+1}} (\beta + \epsilon)^{\frac{n^2+3n+1}{n+1}} x_G^{\frac{n^2+3n+1}{n+1}}. \tag{3.58}$$

Upon simplifying, we obtain the power-law relationship

$$x_G \propto \mathcal{A}^{\frac{n+1}{n(n+2)}}, \tag{3.59}$$

for accumulation rates $\mathcal{A} \gg 1$, which is in agreement with the power law shown in panel (c) of Figures 3.5–3.7. This is the same power law as the $\Lambda \gg H_G$ case in the shear-dominated ice-sheet model.

The models broadly agree on the grounding-line position for large values of slip length and accumulation. This can be explained by returning to Equation (3.54) and directly substituting Equation (3.47). Upon rearranging, we obtain the relationship

$$\mathcal{A}x_G = \left(\frac{n+2}{n+1}\right)^{\frac{n}{n+1}} \Lambda^{\frac{1}{n+1}} H_G^{\frac{n^2+3n+1}{n+1}}, \tag{3.60}$$

which is exactly the flux condition for extensional flow with large accumulation.

3.4.4 Asymptotic solutions for large flotation parameters

As seen in panel (d) of Figures 3.5–3.7, there appears to be a clear power-law relationship between the grounding-line position x_G and the flotation parameter ϵ given by $x_G \propto \epsilon^{-1}$

for both the shear-dominated and extensional ice-sheet models. In Robison et al. [17], it is claimed that the flotation parameter $\epsilon = \beta\frac{\rho_w-\rho}{\rho}$ is typically a small parameter in geophysical scenarios; however, it is illuminating to consider the behaviour of the ice sheet as the flotation parameter ϵ is varied if the bed slope β is small. For our model including accumulation, it is also useful to take the accumulation rate $\mathcal{A}$ as a small parameter, so that $1 + \mathcal{A}x_G \sim 1$. Therefore, in subplot (d) of Figures 3.5–3.7, we take $\mathcal{A} = 10^{-2}$ and $\beta = 10^{-6}$ to better show the power laws for physically relevant values of the flotation parameter.

We derive the power laws theoretically below. To do so, we start from the shear-dominated ice-sheet model, for which the free-surface slope at the grounding line for $\mathcal{A}x_G \ll 1$ is given by

$$\frac{\partial h}{\partial x} = -\left[\frac{1}{H_G^{n+2} + \Lambda H_G^{n+1}}\right]^{1/n}. \tag{3.61}$$

Therefore, the grounding-line position x_G does not appear in the stress condition (3.12a) and so varying the flotation parameter ϵ does not have an effect on the value of the grounding-line depth H_G. The flotation parameter ϵ, therefore, only contributes to the flotation condition at the grounding line, which in the case where $\epsilon \gg \beta$ may be expressed as $x_G = \frac{H_G}{\epsilon}$. Therefore, we obtain the power law

$$x_G \propto \epsilon^{-1}, \tag{3.62}$$

in the limit $\epsilon \gg \beta$, where $\mathcal{A}x_G \ll 1$, which is in agreement with the power law shown in panel (d) of Figures 3.5–3.7.

The power-law relation for the extensional ice-sheet model follows similarly. For $1+\mathcal{A}x_G \sim 1$, the grounding-line flux condition becomes

$$1 = \left(\frac{n+2}{n+1}\right)^{\frac{n}{n+1}} \Lambda^{\frac{1}{n+1}} H_G^{\frac{n^2+3n+1}{n+1}}. \tag{3.63}$$

Therefore, again the flotation parameter ϵ has no effect on the grounding-line depth H_G and only affects the grounding-line position x_G through the flotation condition. Thus, the power law for extensional flow is identical to that of Equation (3.62), which again is in agreement with the power law shown in panel (d) of Figures 3.5–3.7.

Chapter 4

Conclusions

We have explored the flow of marine ice sheets by deriving and solving two models in distinct dynamical regimes. The first describes a regime in which the underside of the ice is only moderately lubricated, in which case the flow is resisted predominantly by vertical shear stress across the depth of the ice sheet. The second regime is one in which the underside of the ice sheet is very well lubricated, where vertical shear is negligible throughout; instead viscous extensional stress becomes dominant. The latter model is relevant to fast-flowing ice streams and shares important dynamical similarities with models of freely floating ice shelves, which also exhibit negligible shear stresses on their upper and lower surfaces.

We couple both of these models of ice sheets, of varying basal lubrication, to a freely floating ice shelf and determine the dominant physical balance at the grounding line, joining the sheet and the shelf. We do so by assuming continuity of thickness, flux and depth-integrated longitudinal stress, which we couple with other geometric conditions at the grounding line. In doing so, we generalise the results of Robison et al. [17], describing shear-dominated ice sheets coupled to an ice shelf, to include the effects of a non-Newtonian shear-thinning rheology of ice as well as basal sliding and ice accumulation.

Having derived the two classes of ice-sheet models from first principles, we examine the resulting dynamics of the flow across a broad range of dimensionless parameter values by solving the full system of governing equations numerically in steady state. A key parameter governing the flow is the dimensionless slip length Λ, which governs the validity of the two models. In particular, the shear-dominated model is valid for $\Lambda \ll 1$ and the extensional

model is valid for $\Lambda \gg 1$. We demonstrate numerically that there is an intermediate range of values of Λ for which the two models agree.

We also characterise power-law relationships appearing for various asymptotic limits, including low and high dimensionless slip lengths Λ, bed slopes β, accumulation rates $\mathcal{A}$, and flotation parameters ϵ. We relate these findings to what we expect on physical grounds.

Several exact and asymptotic results were also obtained. In the no-accumulation limit, a shear-dominated ice sheet has constant depth when bed slope $\beta = 1$, independently of slip length Λ and rheological exponent n. The governing equations for the ice shelf have an exact solution, which approaches a constant which depends on the accumulation rate in the far field. From this, a critical accumulation value can be determined, above which the shelf thickens and below which the shelf thins to its asymptotic limit.

Our parameter sweeps demonstrate that the grounding line position can vary significantly depending on the model used to determine it, demonstrating the importance of empirical knowledge of basal conditions prior to choosing a model. For example, for large bed slopes β, the grounding-line position scales as $\beta^{-(2n+1)/(n+1)}$ and β^{-1} for the shear-dominated and extensional ice-sheet models, respectively. However, in some cases, the models do agree. For example, for sufficiently large slip lengths and shallow bed slopes, the grounding line condition from the shear-dominated ice-sheet model reduces to exactly that of the extensional ice-sheet model. Thus, the two models approach one another for certain parameter regimes, despite being derived from different dominant stress balances.

The models in this thesis have been idealised, restricted to steady, two-dimensional configurations with simplified bed geometry. The calving front has also been mostly neglected. An extension of this work could consider time-dependent solutions, as well as more general bed geometries.

Appendix A

MATLAB Scripts

Here, the MATLAB functions used to solve the governing equations and generate the numerical solutions presented in this thesis are provided. These include the functions used to determine the grounding-line position, solve the shear-dominated and extensional ice-sheet models, and calculate the corresponding velocity profiles in both the grounded ice sheet and floating ice shelf.

A.1 Grounding-Line Functions

A.1.1 Newtonian shear-dominated ice-sheet model

The following MATLAB function determines the grounding-line position for the Newtonian shear-dominated ice-sheet model. It solves the grounding-line stress condition for the positive ice thickness H, using the corresponding expression for the surface gradient h_x. Once H has been obtained, the flotation condition and the linear bed geometry are used to determine x_G, h_G, and b_G.

```
function [xG,H,h,b] = ...
findgroundinglineSheardominatednewtonian( ...
Lambda,beta,Accumulation,epsilon)
%%

x = @(H) H/(beta + epsilon);

hxFun = @(H) -( ...
```

```
    (1 + Accumulation*x(H)) ./ ...
    (H.^3 + Lambda.*H.^2));

F = @(H) ...
    (3/2).*H.^2 ...
    - hxFun(H).^2 ...
    .* ((3/2).*H.^2 + Lambda.*H) ...
    - beta.*Lambda.*hxFun(H).*H ...
    - Accumulation;

%% ----------------------------------------
% Find a positive root only
% ----------------------------------------

Hmin = 1e-10;
Hmax = 1e8;
Hguess = 1;

logHGrid = linspace( ...
    log(Hmin),log(Hmax),2000);

FGrid = arrayfun( ...
    @(logH) F(exp(logH)),logHGrid);

valid = isfinite(FGrid);

idx = find( ...
    valid(1:end-1) ...
    & valid(2:end) ...
    & FGrid(1:end-1).*FGrid(2:end) <= 0);

if isempty(idx)
    error( ...
        ['No positive grounding-line root found ', ...
         'for the Newtonian shear-dominated ', ...
         'ice-sheet model.']);
end
```

```
46
47 % If more than one positive root exists,
48 % choose the one whose bracket is closest
49 % to H = Hguess.
50 logHmid = 0.5.*( ...
51     logHGrid(idx) + logHGrid(idx+1));
52
53 [~,j] = min(abs(logHmid - log(Hguess)));
54 k = idx(j);
55
56 logH = fzero( ...
57     @(logH) F(exp(logH)), ...
58     [logHGrid(k),logHGrid(k+1)]);
59
60 H = exp(logH);
61
62 %% Grounding-line geometry
63
64 xG = H/(beta + epsilon);
65 h = epsilon*xG;
66 b = -beta*xG;
67 end
```

A.1.2 Non-Newtonian shear-dominated ice-sheet model

The following MATLAB function determines the grounding-line position for the non-Newtonian shear-dominated ice-sheet model. First, the surface gradient is expressed in terms of the grounding-line thickness. The grounding-line stress condition is then solved numerically for a positive value of H. Finally, the flotation condition and bed geometry are used to determine the grounding-line position and the corresponding surface and bed elevations.

```
1 function [xG,H,h,b] = ...
2 findgroundinglineSheardominated( ...
3 Lambda,beta,Accumulation,n,epsilon)
4 %%
5
```

```
x = @(H) H/(beta + epsilon);

slipCoeff = Lambda;

hxFun = @(H) -( ...
    (1 + Accumulation*x(H)) ./ ...
    (H.^(n+2) ...
    + slipCoeff.*H.^(n+1)) ...
    ).^(1/n);

F = @(H) ...
    Accumulation ...
    + hxFun(H).*abs(hxFun(H)).^(n-1) ...
    .* ( ...
        (n+2)/(n+1).*H.^(n+1).*hxFun(H) ...
        + slipCoeff.*H.^n ...
        .* (hxFun(H) + beta)) ...
    - (n+2)/(n+1).*H.^(n+1);

%% ----------------------------------------
% Find a positive root only
% ----------------------------------------

Hmin = 1e-10;
Hmax = 1e8;
Hguess = 1;

logHGrid = linspace( ...
    log(Hmin),log(Hmax),2000);

FGrid = arrayfun( ...
    @(logH) F(exp(logH)),logHGrid);

valid = isfinite(FGrid);

idx = find( ...
    valid(1:end-1) ...
```

```
    & valid(2:end) ...
    & FGrid(1:end-1).*FGrid(2:end) <= 0);

if isempty(idx)
    error( ...
        ['No positive grounding-line root found ', ...
         'for the shear-dominated ice-sheet model.']);
end

logHmid = 0.5.*( ...
    logHGrid(idx) + logHGrid(idx+1));

[~,j] = min(abs(logHmid - log(Hguess)));
k = idx(j);

logH = fzero( ...
    @(logH) F(exp(logH)), ...
    [logHGrid(k),logHGrid(k+1)]);

H = exp(logH);

% Apply grounding-line geometry from
% flotation and b = -beta*x.
xG = H/(beta + epsilon);
h = epsilon*xG;
b = -beta*xG;
end
```

A.1.3 Extensional ice-sheet model

The following MATLAB function determines the grounding-line position for the extensional ice-sheet model. The grounding-line condition is reduced to a non-linear equation for the ice thickness H, which is solved numerically subject to the requirement that the solution is positive. The corresponding grounding-line position and elevations are then calculated using flotation and the prescribed linear bed.

```
function [xG,H,h,b] = ...
```

```
findgroundinglineExtensional( ...
Lambda,beta,Accumulation,n,epsilon)

if Lambda <= 0
    error( ...
        ['Lambda must be positive for ', ...
         'extensional ice-sheet model.']);
end

p = (n^2 + 3*n + 1)/(n + 1);

K = ((n+2)/(n+1))^(n/(n+1)) ...
    * Lambda^(1/(n+1));

F = @(H) ...
    1 ...
    + Accumulation*H/(beta + epsilon) ...
    - K*H.^p;

%% ----------------------------------------
% Find a positive root only
% ----------------------------------------

Hmin = 1e-10;
Hmax = 1e8;

logHGrid = linspace( ...
    log(Hmin),log(Hmax),2000);

FGrid = arrayfun( ...
    @(logH) F(exp(logH)),logHGrid);

valid = isfinite(FGrid);

idx = find( ...
    valid(1:end-1) ...
    & valid(2:end) ...
```

```
    & FGrid(1:end-1).*FGrid(2:end) <= 0, ...
    1,'first');

if isempty(idx)
    error( ...
        ['No positive grounding-line root found ', ...
         'for the extensional ice-sheet model.']);
end

logH = fzero( ...
    @(logH) F(exp(logH)), ...
    [logHGrid(idx),logHGrid(idx+1)]);

H = exp(logH);

%% Grounding-line geometry

xG = H/(beta + epsilon);
h = epsilon*xG;
b = -beta*xG;
end
```

A.2 Ice-Sheet Solvers

A.2.1 Newtonian shear-dominated ice-sheet model

The following MATLAB function constructs the complete steady ice-sheet and ice-shelf profile for the Newtonian shear-dominated ice-sheet model. The grounding-line quantities are first calculated, after which the sheet equation is integrated upstream from the grounding line to the divide and the shelf equation is integrated downstream. The surface and bed elevations are then reconstructed from the ice thickness and the appropriate grounded and floating geometry.

```
function [ ...
xsheet,xshelf,hsheet,hshelf, ...
bsheet,bshelf,hxsheet,xG,HG,hG,bG] = ...
```

```
Sheardominatedicesheetnewtonian( ...
Lambda,beta,Accumulation, ...
epsilon,shelfextent)
%%
% Sheardominatedicesheetnewtonian solves the
% governing equations for the Newtonian
% shear-dominated ice-sheet model.
%
% It returns h and b for both the sheet and shelf,
% the x values in both regions, and the grounding-
% line quantities xG, HG, hG and bG.

% Find grounding-line position.
[xG,HG,hG,bG] = ...
findgroundinglineSheardominatednewtonian( ...
Lambda,beta,Accumulation,epsilon);

% Solve sheet equations upstream from x_G to 0.
[xsheet,Hsheet] = ode45( ...
@(x,H) sheetequationSheardominatednewtonian( ...
x,H,Lambda,beta,Accumulation), ...
[xG,0],HG);

% Apply b = -Ax and h = H+b.
bsheet = -beta*xsheet;
hsheet = Hsheet + bsheet;

% h_x = H_x - beta, used to determine velocity.
hxsheet = -( ...
(1 + Accumulation*xsheet) ./ ...
(Hsheet.^3 + Lambda.*Hsheet.^2));

% Solve shelf equations downstream from x_G.
[xshelf,Hshelf] = ode45( ...
@(x,H) shelfequationSheardominatednewtonian( ...
x,H,Accumulation), ...
[xG,xG*shelfextent],HG);
```

```
41
42 % Use flotation condition to determine h and b.
43 flotationh = epsilon/(beta + epsilon);
44 flotationb = beta/(beta + epsilon);
45
46 hshelf = flotationh*Hshelf;
47 bshelf = -flotationb*Hshelf;
48
49 end
50
51 function dHdx = ...
52 sheetequationSheardominatednewtonian( ...
53 x,H,Lambda,beta,Accumulation)
54 % Solve the Newtonian shear-dominated
55 % ice-sheet model sheet-depth equation.
56
57 dHdx = ...
58     beta ...
59     - (1 + Accumulation*x) ./ ...
60     (H.^3 + Lambda.*H.^2);
61 end
62
63 function dHdx = ...
64 shelfequationSheardominatednewtonian( ...
65 x,H,Accumulation)
66 % Solve the Newtonian shear-dominated
67 % ice-sheet model shelf-depth equation.
68
69 dHdx = ...
70     (Accumulation*H - (3/2).*H.^3) ...
71     ./ (1 + Accumulation*x);
72 end
```

A.2.2 Non-Newtonian shear-dominated ice-sheet model

The following MATLAB function constructs the complete steady ice-sheet and ice-shelf profile for the non-Newtonian shear-dominated ice-sheet model. After locating the grounding

line, the non-Newtonian sheet equation is integrated upstream and the shelf equation is integrated downstream. The resulting ice thickness is then used to determine the surface and bed profiles in each region.

```
function [ ...
xsheet,xshelf,hsheet,hshelf, ...
bsheet,bshelf,hxsheet,xG,HG,hG,bG] = ...
Sheardominatedicesheet( ...
Lambda,beta,Accumulation,n, ...
epsilon,shelfextent)
%%
% Sheardominatedicesheet solves the governing
% equations for the non-Newtonian shear-dominated
% ice-sheet model.
%
% It returns h and b for both the sheet and shelf,
% the x values in both regions, and the grounding-
% line quantities xG, HG, hG and bG.

% Find grounding-line position.
[xG,HG,hG,bG] = ...
findgroundinglineSheardominated( ...
Lambda,beta,Accumulation,n,epsilon);

% Solve sheet equations upstream from x_G to 0.
[xsheet,Hsheet] = ode45( ...
@(x,H) sheetequationSheardominated( ...
x,H,Lambda,beta,Accumulation,n), ...
[xG,0],HG);

% Apply b = -Ax and h = H+b.
bsheet = -beta*xsheet;
hsheet = Hsheet + bsheet;

% h_x = H_x - beta, used to determine velocity.
hxsheet = -( ...
(1 + Accumulation*xsheet) ./ ...
(Hsheet.^(n+2) ...
```

```
+ Lambda.*Hsheet.^(n+1)) ...
).^(1/n);

% Solve shelf equations downstream from x_G.
[xshelf,Hshelf] = ode45( ...
@(x,H) shelfequationSheardominated( ...
x,H,Accumulation,n), ...
[xG,xG*shelfextent],HG);

% Use flotation condition to determine h and b.
flotationh = epsilon/(beta + epsilon);
flotationb = beta/(beta + epsilon);

hshelf = flotationh*Hshelf;
bshelf = -flotationb*Hshelf;

end

function dHdx = ...
sheetequationSheardominated( ...
x,H,Lambda,beta,Accumulation,n)
% Solve the non-Newtonian shear-dominated
% ice-sheet model sheet-depth equation.

dHdx = ...
    beta ...
    - ( ...
        (1 + Accumulation*x) ./ ...
        (H.^(n+2) ...
        + Lambda.*H.^(n+1)) ...
      ).^(1/n);
end

function dHdx = ...
shelfequationSheardominated( ...
x,H,Accumulation,n)
% Solve the non-Newtonian shear-dominated
```

```
% ice-sheet model shelf-depth equation.

dHdx = ...
    (Accumulation*H ...
    - (n+2)/(n+1).*H.^(n+2)) ...
    ./ (1 + Accumulation*x);
end
```

A.2.3 Extensional ice-sheet model

The following MATLAB function constructs the steady ice-sheet and ice-shelf profile for the extensional ice-sheet model. The grounding line is first determined from the extensional ice-sheet grounding-line condition. The corresponding sheet equation is then integrated upstream and the shelf equation downstream, with the bed and surface elevations reconstructed from the resulting ice thickness.

```
function [ ...
xsheet,xshelf,hsheet,hshelf, ...
bsheet,bshelf,xG,HG,hG,bG] = ...
Extensionalicesheet( ...
Lambda,beta,Accumulation,n, ...
epsilon,shelfextent)
%%
% Extensionalicesheet solves the governing
% equations for the non-Newtonian extensional
% ice-sheet model.
%
% It returns h and b for both the sheet and shelf,
% the x values in both regions, and the grounding-
% line quantities xG, HG, hG and bG.

% Find grounding-line position.
[xG,HG,hG,bG] = ...
findgroundinglineExtensional( ...
Lambda,beta,Accumulation,n,epsilon);

% Solve sheet equations upstream from x_G to 0.
```

```
[xsheet,Hsheet] = ode45( ...
@(x,H) sheetequationExtensional( ...
x,H,Lambda,beta,Accumulation,n), ...
[xG,0],HG);

% Apply b = -Ax and h = H+b.
bsheet = -beta*xsheet;
hsheet = Hsheet + bsheet;

% Solve shelf equations downstream from x_G.
[xshelf,Hshelf] = ode45( ...
@(x,H) shelfequationExtensional( ...
x,H,Accumulation,n), ...
[xG,xG*shelfextent],HG);

% Use flotation condition to determine h and b.
flotationh = epsilon/(beta + epsilon);
flotationb = beta/(beta + epsilon);

hshelf = flotationh*Hshelf;
bshelf = -flotationb*Hshelf;

end

function dHdx = ...
sheetequationExtensional( ...
x,H,Lambda,beta,Accumulation,n)
% Solve the non-Newtonian extensional
% ice-sheet model sheet-depth equation.

dHdx = ...
    beta ...
    - ( ...
        (1 + Accumulation*x) ./ ...
        (Lambda.*H.^(n+1)) ...
      ).^(1/n);
end
```

```
59
60 function dHdx = ...
61 shelfequationExtensional( ...
62 x,H,Accumulation,n)
63 % Solve the extensional ice-sheet
64 % shelf-depth equation.
65
66 dHdx = ...
67     (Accumulation*H ...
68     - (n+2)/(n+1).*H.^(n+2)) ...
69     ./ (1 + Accumulation*x);
70 end
```

A.2.4 Constant-shelf-depth shear-dominated model

The function `SheardominatedicesheetConstantShelf.m` constructs the steady ice-sheet and ice-shelf profile for the shear-dominated ice-sheet model subject to the additional requirement that the shelf depth remains constant. The accumulation rate required to satisfy this condition is determined together with the grounding-line thickness and position. The grounded sheet and floating shelf equations are then integrated from the grounding line, and the corresponding surface and bed elevations are reconstructed.

```
1  function [ xsheet,xshelf,hsheet,hshelf, bsheet,bshelf,hxsheet,xG,
       ...
2      HG,hG,bG,Accumulation] = SheardominatedicesheetConstantShelf(
           ...
3          Lambda,beta,n, epsilon,shelfextent)
4  %%
5  % SheardominatedicesheetConstantShelf solves the governing
       equations
6  % for the shear-dominated ice-sheet model subject to constant shelf
       depth.
7  % It returns h and b for both the sheet and shelf,
8  % the x values in both regions, the sheet surface gradient,
9  % the grounding-line quantities xG, HG, hG and bG,
10 % and the accumulation rate required for constant shelf depth.
11
```

```
12 % Find grounding-line position and accumulation rate.
13 [xG,HG,hG,bG,Accumulation] = ...
14     findgroundinglineConstantShelfSheardominated( Lambda,beta,n,
       epsilon);
15
16 % Solve sheet equations upstream from x_G to 0.
17 [xsheet,Hsheet] = ode45( @(x,H) sheetequationSheardominated( ...
18         x,H,Lambda,beta,Accumulation,n), [xG,0], HG);
19
20 % Apply b = -beta*x and h = H+b.
21 bsheet = -beta*xsheet;
22 hsheet = Hsheet + bsheet;
23
24 % Surface gradient h_x.
25 hxsheet = -( (1 + Accumulation*xsheet) ./ ( Hsheet.^(n+2) ...
26         + Lambda.*Hsheet.^(n+1) ) ).^(1/n);
27
28 % Solve shelf equations downstream from x_G.
29 [xshelf,Hshelf] = ode45( @(x,H) shelfequationSheardominated( ...
30         x,H,Accumulation,n), [xG,xG*shelfextent], HG);
31
32 % Use flotation condition to determine shelf h and b.
33 flotationh = epsilon/(beta + epsilon);
34 flotationb = beta/(beta + epsilon);
35
36 hshelf = flotationh*Hshelf;
37 bshelf = -flotationb*Hshelf;
38
39 end
40
41 function dHdx = sheetequationSheardominated( ...
42         x,H,Lambda,beta,Accumulation,n)
43 % Solve the shear-dominated ice-sheet depth equation.
44
45     dHdx = beta - ( (1 + Accumulation*x) ./ ( H.^(n+2) ...
46                 + Lambda.*H.^(n+1) ) ).^(1/n);
47
```

```
48 end
49
50 function dHdx = shelfequationSheardominated( x,H,Accumulation,n)
51 % Solve the ice-shelf depth equation.
52
53     dHdx = ( Accumulation*H - (n+2)/(n+1) .* H.^(n+2) ) ./ ...
54         (1 + Accumulation*x);
55
56 end
57
58 function [xG,H,h,b,Accumulation] = ...
59     findgroundinglineConstantShelfSheardominated( Lambda,beta,n,
       epsilon)
60 %%
61 % Determine the grounding-line depth for a constant-depth shelf.
62
63     x = @(H) H/(beta + epsilon);
64
65     % Constant-shelf-depth accumulation condition.
66     % Accumulation = (n+2)/(n+1) H_G^(n+1).
67     accumulationFun = @(H) (n+2)/(n+1) .* H.^(n+1);
68
69     % Surface gradient at the grounding line.
70     hxFun = @(H) -( ( 1 + accumulationFun(H).*x(H) ) ./ ( H.^(n+2)
       ...
71          + Lambda.*H.^(n+1) ) ).^(1/n);
72
73     % Full grounding-line stress condition.
74     F = @(H) accumulationFun(H) + hxFun(H) .* abs(hxFun(H)).^(n-1)
       ...
75         .* ( (n+2)/(n+1) .* H.^(n+1) .* hxFun(H) + Lambda .* H.^n
           ...
76             .* (hxFun(H) + beta) ) - (n+2)/(n+1) .* H.^(n+1);
77
78     %% Find a positive grounding-line thickness
79
80     Hmin = 1e-10;
```

```
    Hmax = 1e8;
    Hguess = 1;

    logHGrid = linspace( log(Hmin), log(Hmax), 2000);

    FGrid = arrayfun( @(logH) F(exp(logH)), logHGrid);

    valid = isfinite(FGrid);

    idx = find( valid(1:end-1) & valid(2:end) & FGrid(1:end-1) ...
        .* FGrid(2:end) <= 0);

    if isempty(idx)
        error( ['No positive constant-shelf shear-dominated ', ...
            'ice-sheet grounding-line root found.']);
    end

    logHmid = 0.5.*( logHGrid(idx) + logHGrid(idx+1));

    [~,j] = min( abs(logHmid - log(Hguess)));

    k = idx(j);

    logH = fzero( @(logH) F(exp(logH)), [ logHGrid(k), logHGrid(k
       +1) ]);

    H = exp(logH);

    %% Grounding-line geometry and accumulation

    Accumulation = accumulationFun(H);

    xG = H/(beta + epsilon);
    h = epsilon*xG;
    b = -beta*xG;

end
```

A.2.5 Constant-shelf-depth extensional model

The function `ExtensionalicesheetConstantShelf.m` constructs the corresponding steady solution for the extensional ice-sheet model when the shelf depth is constrained to remain constant. The constant-shelf-depth condition determines the accumulation rate as a function of the grounding-line thickness, while the extensional grounding-line condition is used to locate the grounding line. The sheet and shelf equations are subsequently integrated upstream and downstream, respectively.

```
function [ xsheet,xshelf,hsheet,hshelf, ...
    bsheet,bshelf,xG,HG,hG,bG,Accumulation] = ...
    ExtensionalicesheetConstantShelf( Lambda,beta,n, epsilon,
        shelfextent)
%%
% ExtensionalicesheetConstantShelf solves the governing equations
% for the extensional ice-sheet model subject to constant shelf
    depth.

% Determine the grounding-line position and accumulation rate.
[xG,HG,hG,bG,Accumulation] = ...
    findgroundinglineConstantShelfExtensional( Lambda,beta,n,
        epsilon);

% Solve sheet equations upstream from x_G to 0.
[xsheet,Hsheet] = ode45( @(x,H) sheetequationExtensional( ...
        x,H,Lambda,beta,Accumulation,n), [xG,0], HG);

% Apply b = -beta*x and h = H+b.
bsheet = -beta*xsheet;
hsheet = Hsheet + bsheet;

% Solve shelf equations downstream from x_G.
[xshelf,Hshelf] = ode45( @(x,H) shelfequationExtensional( ...
        x,H,Accumulation,n), [xG,xG*shelfextent], HG);

% Apply the flotation condition in the shelf.
flotationh = epsilon/(beta + epsilon);
flotationb = beta/(beta + epsilon);
```

```
27
28 hshelf = flotationh*Hshelf;
29 bshelf = -flotationb*Hshelf;
30
31 end
32
33 function dHdx = sheetequationExtensional( x,H,Lambda,beta,
       Accumulation,n)
34
35     dHdx = beta - ( (1 + Accumulation*x) ./ ( Lambda.*H.^(n+1) ) )
           .^(1/n);
36
37 end
38
39 function dHdx = shelfequationExtensional( x,H,Accumulation,n)
40
41     dHdx = ( Accumulation*H - (n+2)/(n+1) .* H.^(n+2) ) ./ ...
42         (1 + Accumulation*x);
43
44 end
45
46 function [xG,H,h,b,Accumulation] = ...
47     findgroundinglineConstantShelfExtensional( Lambda,beta,n,
           epsilon)
48
49     if Lambda <= 0
50         error( ['Lambda must be positive ', ...
51               'for the extensional ice-sheet model.']);
52     end
53
54     x = @(H) H/(beta + epsilon);
55
56     accumulationFun = @(H) (n+2)/(n+1) .* H.^(n+1);
57
58     p = (n^2 + 3*n + 1) /(n + 1);
59
60     K = ((n+2)/(n+1)) ^(n/(n+1)) .* Lambda ^(1/(n+1));
```

```
61
62      % Extensional grounding-line flux condition.
63      F = @(H) 1 + accumulationFun(H).*x(H) - K.*H.^p;
64
65      %% Find a positive grounding-line thickness
66
67      Hmin = 1e-10;
68      Hmax = 1e8;
69      Hguess = 1;
70
71      logHGrid = linspace( log(Hmin), log(Hmax), 2000);
72
73      FGrid = arrayfun( @(logH) F(exp(logH)), logHGrid);
74
75      valid = isfinite(FGrid);
76
77      idx = find( valid(1:end-1) & valid(2:end) & FGrid(1:end-1) ...
78          .* FGrid(2:end) <= 0);
79
80      if isempty(idx)
81          error( ['No positive constant-shelf extensional ', ...
82                  'ice-sheet grounding-line root found.']);
83      end
84
85      logHmid = 0.5.*( logHGrid(idx) + logHGrid(idx+1));
86
87      [~,j] = min( abs(logHmid - log(Hguess)));
88
89      k = idx(j);
90
91      logH = fzero( @(logH) F(exp(logH)), [ logHGrid(k), logHGrid(k
            +1) ]);
92
93      H = exp(logH);
94
95      %% Grounding-line geometry and accumulation
96
```

```
97      Accumulation = accumulationFun(H);
98
99      xG = H/(beta + epsilon);
100     h = epsilon*xG;
101     b = -beta*xG;
102
103 end
```

A.3 Velocity Profiles

A.3.1 Newtonian shear-dominated ice-sheet model

The following MATLAB function evaluates the vertical velocity profile within the grounded sheet for the Newtonian shear-dominated ice-sheet model. The velocity is written as the sum of a basal sliding contribution and a depth-dependent deformation contribution, with the vertical coordinate η scaled such that $\eta = 0$ at the bed and $\eta = 1$ at the surface.

```
1  function u = ...
2  velocityprofileSheardominatedsheetnewtonian( ...
3  eta,H,hx,Lambda)
4  %VELOCITYPROFILESHEARDOMINATEDSHEETNEWTONIAN
5  %
6  % Nondimensional Newtonian shear-dominated
7  % ice-sheet model velocity profile.
8  %
9  % eta = 0 at the bed
10 % eta = 1 at the surface
11 %
12 % hx = dh/dx < 0
13
14 s = -hx;
15 beta = 3/2;
16
17 % Basal sliding velocity.
18 ub = Lambda.*H.*s;
19
20 % Full velocity profile.
```

```
u = ...
    ub ...
    + beta.*H.^2.*s ...
    .* (1 - (1-eta).^2);
end
```

A.3.2 Non-Newtonian shear-dominated ice-sheet model

The following MATLAB function evaluates the vertical velocity profile within the grounded sheet for the non-Newtonian shear-dominated ice-sheet model. As in the Newtonian case, the velocity consists of a basal sliding component together with a depth-dependent deformation component. The dependence of both contributions on the surface gradient and ice thickness is modified by the non-Newtonian exponent n.

```
function u = ...
velocityprofileSheardominatedsheet( ...
eta,H,hx,Lambda,n)
%VELOCITYPROFILESHEARDOMINATEDSHEET
%
% Nondimensional shear-dominated
% ice-sheet model velocity profile.
%
% eta = 0 at the bed
% eta = 1 at the surface
%
% hx = dh/dx < 0

s = -hx;
beta = (n+2)/(n+1);

% Basal sliding velocity.
ub = Lambda.*H.^n.*s.^n;

% Full velocity profile.
u = ...
    ub ...
    + beta.*H.^(n+1).*s.^n ...
```

```
    .* (1 - (1-eta).^(n+1));
end
```

A.3.3 Extensional ice-sheet model

The following MATLAB function evaluates the velocity within the grounded region for the extensional ice-sheet model. In this approximation the velocity is taken to be independent of depth and is therefore determined directly from the local flux divided by the ice thickness. The coordinate η is retained only so that the function has the same input structure as the shear-dominated ice-sheet velocity-profile functions.

```
function u = ...
velocityprofileExtensionalsheet( ...
eta,x,H,Accumulation)
%VELOCITYPROFILEEXTENSIONALSHEET
%
% Extensional ice-sheet model outer solution.
% Velocity is independent of depth.
%
% eta is included so that this function
% has the same style as the shear-dominated
% ice-sheet model velocity-profile function.

q = 1 + Accumulation*x;
u0 = q./H;

u = u0.*ones(size(eta));
end
```

A.3.4 The ice shelf

The following MATLAB function evaluates the velocity within the floating ice shelf. The shelf velocity is assumed to be independent of depth and is obtained from the local ice flux divided by the shelf thickness. Consequently, the same velocity is returned at every vertical position η.

```
function u = ...
```

```
velocityprofileshelf( ...
eta,x,H,Accumulation)
%VELOCITYPROFILESHELF
%
% Depth-independent shelf velocity.

q = 1 + Accumulation*x;
u0 = q./H;

u = u0.*ones(size(eta));
end
```

References

[1] Richard B. Alley, Sridhar Anandakrishnan, Todd K. Dupont, Byron R. Parizek, and David Pollard. Effect of sedimentation on ice-sheet grounding-line stability. *Science*, 315(5820):1838–1841, 2007.

[2] Sridhar Anandakrishnan, Ginny A. Catania, Richard B. Alley, and Huw J. Horgan. Discovery of till deposition at the grounding line of Whillans Ice Stream. *Science*, 315(5820):1835–1838, 2007.

[3] Matthew R. Bennett. Ice streams as the arteries of an ice sheet: their mechanics, stability and significance. *Earth-Science Reviews*, 61(3):309–339, 2003.

[4] Robert G. Bingham, Martin J. Siegert, D. A. Young, and D. D. Blankenship. Organized flow from the South Pole to the Filchner–Ronne Ice Shelf: an assessment of balance velocities in interior East Antarctica using radio echo sounding data. *Journal of Geophysical Research: Earth Surface*, 112:F03S26, 2007.

[5] Julien Brondex, Olivier Gagliardini, Fabien Gillet-Chaulet, and Gaël Durand. Sensitivity of grounding line dynamics to the choice of the friction law. *Journal of Glaciology*, 63(241):854–866, 2017.

[6] A. C. Fowler. Sliding with cavity formation. *Journal of Glaciology*, 33(115):255–267, 1987.

[7] A. C. Fowler and D. A. Larson. On the flow of polythermal glaciers. 1. model and preliminary analysis. *Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences*, 363(1713):217–242, 1978.

[8] J. W. Glen. The creep of polycrystalline ice. *Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences*, 228(1175):519–538, 1955.

[9] Ralf Greve. Application of a polythermal three-dimensional ice sheet model to the Greenland Ice Sheet: response to steady-state and transient climate scenarios. *Journal of Climate*, 10(5):901–918, 1997.

[10] Ralf Greve and Heinz Blatter. *Dynamics of Ice Sheets and Glaciers*. Springer, Berlin, Heidelberg, 2009.

[11] Philippe Huybrechts. The Antarctic Ice Sheet during the last glacial-interglacial cycle: a three-dimensional experiment. *Annals of Glaciology*, 14:115–119, 1990.

[12] Katarzyna N. Kowal, Samuel S. Pegler, and M. Grae Worster. Dynamics of laterally confined marine ice sheets. *Journal of Fluid Mechanics*, 790:R2, 2016.

[13] Douglas R. MacAyeal. Large-scale ice flow over a viscous basal sediment: theory and application to Ice Stream B, Antarctica. *Journal of Geophysical Research: Solid Earth*, 94(B4):4071–4087, 1989.

[14] Isabelle Muszynski and G. E. Birchfield. A coupled marine ice-stream–ice-shelf model. *Journal of Glaciology*, 33(113):3–15, 1987.

[15] J. F. Nye. A calculation on the sliding of ice over a wavy surface using a newtonian viscous approximation. *Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences*, 311(1506):445–467, 1969.

[16] Samuel S. Pegler, Katarzyna N. Kowal, Leonard Q. Hasenclever, and M. Grae Worster. Lateral controls on grounding-line dynamics. *Journal of Fluid Mechanics*, 722:R1, 2013.

[17] Rosalyn A. V. Robison, Herbert E. Huppert, and M. Grae Worster. Dynamics of viscous grounding lines. *Journal of Fluid Mechanics*, 648:363–380, 2010.

[18] Jonathan P. Rothstein. Slip on superhydrophobic surfaces. *Annual Review of Fluid Mechanics*, 42:89–109, 2010.

[19] Roiy Sayag and M. Grae Worster. Axisymmetric gravity currents of power-law fluids over a rigid horizontal surface. *Journal of Fluid Mechanics*, 716:R5, 2013.

[20] Christian Schoof. The effect of cavitation on glacier sliding. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 461(2055):609–627, 2005.

[21] Christian Schoof. Marine ice-sheet dynamics. part 1. the case of rapid sliding. *Journal of Fluid Mechanics*, 573:27–55, 2007.

[22] Christian Schoof and Ian Hewitt. Ice-sheet dynamics. *Annual Review of Fluid Mechanics*, 45:217–239, 2013.

[23] Robert H. Thomas. Ice shelves: a review. *Journal of Glaciology*, 24(90):273–286, 1979.

[24] Martin Truffer and Keith A. Echelmeyer. Of isbræ and ice streams. *Annals of Glaciology*, 36:66–72, 2003.

[25] Katarzyna L. P. Warburton, Duncan R. Hewitt, Colin R. Meyer, and Jerome A. Neufeld. A shallow approximation for ice streams sliding over strong beds. *Journal of Glaciology*, 69(278):1634–1645, 2023.

[26] J. Weertman. On the sliding of glaciers. *Journal of Glaciology*, 3(21):33–38, 1957.

[27] Zihan Yan and Katarzyna N. Kowal. A controllable sliding law for thin-film flows over slippery fluid-saturated substrates: theory and experiments. *Journal of Fluid Mechanics*, 982:A14, 2024.